

Traceability Matrix

REMEMBER GORDON! – Rulebook $\rightleftharpoons$ Implementation Mapping

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Overview

Implemented	Descriptive	Implicit	Out-of-scope
83	28	4	9

Total mappings: 124 · Total impl sites: 246

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0/3 implemented

§1 – Introduction

§1 – Introduction

descriptive

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Introduction

§1.1 – General Comments

descriptive

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General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior

(see manual for full text)

§1.2 – Game Scale

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Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

 3/7 implemented

§2 – Game Components

§2 – Game Components

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Game Components

§2.1 – The Game Maps

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The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

§2.2 – Play Aids

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Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

§2.3 – The Units

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The Units

File	Symbol	Code Snippet
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omdurman-types/src/lib.rs:864 GH:omdurman-types/src/lib.rs:864	UnitKind	<pre> 862 /// `Some(UnitKind::Marker)` or `None`. 863 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 864 pub enum UnitKind { 865 /// Foot infantry (\$2.3): fire / melee / movement. 866 Infantry { </pre>
omdurman-rules/src/lib.rs:744 GH:omdurman-rules/src/lib.rs:744	UnitProfile	<pre> 742 /// print no melee value). 743 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 744 pub struct UnitProfile { 745 pub kind: UnitKind, 746 pub identity: UnitIdentity, </pre>
omdurman-rules/src/lib.rs:16 GH:omdurman-rules/src/lib.rs:16	BrigadeId	<pre> 14 15 use omdurman_types::{ 16 BrigadeId, BrigadeNationality, DayNight, DervishTribe, Faction, HexCoord, HexsideRef, Player, 17 SetupLetter, UnitKind, 18 }; </pre>
omdurman-types/src/lib.rs:679 GH:omdurman-types/src/lib.rs:679	SpriteAnnotation	<pre> 677 /// as an optional overlay over the compiled `sprite_data` fallback. 678 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)] 679 pub struct SpriteAnnotation { 680 pub color: SpriteColor, 681 pub faction: Option<Faction>, </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::british_army_row_zero_specials_classify_by_counter](#)
[omdurman-rules::src::unit_profiles::egyptian_army_row_zero_specials_classify_by_counter](#)

§2.4 – Game Parts Inventory

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Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

§2.31 – Dervish weapon types

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Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:476	WeaponClass	<pre> 474 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 475)] 476 pub enum WeaponClass { </pre>

GH:omdurman-rules/src/lib.rs:476		<pre> 477 /// Dervish spears and swords -- no ranged fire at all. 478 Melee,</pre>
omdurman-rules/src/unit_profiles.rs:510 GH:omdurman-rules/src/unit_profiles.rs:510	dervish_tribe	<pre> 508 /// Resolve a Dervish tribal foot counter (§2.31): Jehadia, Danagla and 509 /// Isa Zachneih fire on the rifles line; every other tribe is spear-armed. 510 pub fn dervish_tribe(tribe: DervishTribe) -> Option<Classification> { 511 // §2.31: "Jehadia and Danagla units fire on the 'rifles' line as does the 512 // Isa Zachneih unit. All other Dervish units (including leaders) are armed</pre>
omdurman-rules/src/unit_profiles.rs:292 GH:omdurman-rules/src/unit_profiles.rs:292	khalifa_abdullah	<pre> 290 /// battle (§9.322). All three are interchangeable, so they share the 291 /// `DervishArtillery` identity. 292 pub fn khalifa_abdullah(col: u32, row: u32) -> Option<Classification> { 293 let artillery = { 294 Some(Classification {</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::dervish_weapon_class_follows_the_rifles_line](#)

§2.32 – Anglo-Egyptian weapon types

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

****Sample Dervish Units**** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

****Sample Anglo-Egyptian Units**** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:428 GH:omdurman-rules/src/lib.rs:428	GunboatId	<pre> 426 /// fire; "old" gunboats do not (rulebook §2.32). 427 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 428 pub enum GunboatId { 429 /// One of the five new-type named gunboats with howitzer capability. 430 Named(NamedGunboat),</pre>
omdurman-rules/src/lib.rs:447 GH:omdurman-rules/src/lib.rs:447	NamedGunboat	<pre> 445 /// The five named gunboats with howitzer capability (rulebook §6.64, §2.32). 446 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 447 pub enum NamedGunboat { 448 Sultan, 449 Melik,</pre>
omdurman-rules/src/lib.rs:460	OldGunboat	<pre> 458 /// in the Maxim Second Fire and Howitzer subphase (§6.42). 459 #[derive(Serialize, Deserialize, Clone, Copy,</pre>

GH:omdurman-rules/src/lib.rs:460		<pre>PartialEq, Eq, Hash, Debug, strum::Display]] 460 pub enum OldGunboat { 461 LordKitchener, 462 Tamai,</pre>
omdurman-rules/src/lib.rs:460 GH:omdurman-rules/src/lib.rs:460	GunboatId::Old	<pre>458 /// in the Maxim Second Fire and Howitzer subphase 459 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 460 pub enum OldGunboat { 461 LordKitchener, 462 Tamai,</pre>
omdurman-rules/src/lib.rs:434 GH:omdurman-rules/src/lib.rs:434	GunboatId::Dervish-Gunboat	<pre>432 Old(OldGunboat), 433 /// A Dervish gunboat (§9.111, §10.14). 434 DervishGunboat(u8), 435 } 436 </pre>

Covered by tests: [omdurman-rules::src::effects::tests::old_gunboat_lacks_howitzer](#)
[omdurman-rules::src::effects::tests::old_gunboat_rejected_from_howitzer_subphase](#)

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| §3 – Getting Started

§3 – Getting Started

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Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

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§4 – Turn Sequence

§4 – Turn Sequence

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Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
 1. Direct Fire Subphase
 2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:252 GH:omdurman-rules/src/lib.rs:252	GameTurnIndex	<pre>250 /// One-based Game Turn index (1, 2, ... up to the 251 scenario length) (rulebook §4). 252 #[derive(Serialize, Deserialize, Clone, Copy, 253 PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 254 pub struct GameTurnIndex(u8); 255 256 impl GameTurnIndex {</pre>
omdurman-rules/src/lib.rs:274 GH:omdurman-rules/src/lib.rs:274	Phase	<pre>272 /// etc. 273 #[derive(Serialize, Deserialize, Clone, Copy, 274 PartialEq, Eq, Debug, Default)] 275 pub enum Phase { 276 /// Pre-game deployment (§9.2/§9.3/§10): fixed 277 units are placed, each side 278 /// deploys its order of battle within its legal 279 zone, and river</pre>
omdurman-rules/src/effects/state.rs:5 GH:omdurman-rules/src/effects/state.rs:5	GameState	<pre>3 /// All mutable state of a game in progress (rulebook 4 §4). 5 #[derive(Serialize, Deserialize, Clone, Debug)] 6 pub struct GameState { 7 pub scenario: Scenario, 8 pub current_turn: GameTurnIndex,</pre>
omdurman-rules/src/effects/state.rs:141 GH:omdurman-rules/src/effects/state.rs:141	GameState::new	<pre>139 impl GameState { 140 /// Create a fresh game state for a given scenario 141 (rulebook §4). 142 pub fn new(scenario: Scenario) -> Self { 143 let first = scenario_turn(scenario, 144 GameTurnIndex::new(1)); 145 // First player to *move* per scenario: 146 Campaign -- Anglo-Egyptian moves</pre>
omdurman-rules/src/effects/effect.rs:17 GH:omdurman-rules/src/effects/effect.rs:17	AdvancePhase	<pre>15 /// At end-of-turn, disrupted units recover and 16 per-turn tracking is 17 /// cleared. 18 AdvancePhase, 19 // -- Movement 20 -----</pre>
omdurman-rules/src/effects/dispatch.rs:98 GH:omdurman-	advance_phase	<pre>96 97 /// Advance the game state to the next phase (rulebook 98 §4).</pre>

rules/src/effects/ dispatch.rs:98		<pre>98 pub fn advance_phase(state: &mut GameState) -> Result<(), RuleError> { 99 let old_phase = state.phase; 100 debug!(old_phase = ?old_phase, active_player = ? state.active_player, "advance_phase");</pre>
omdurman- rules/src/effects/ dispatch.rs:202 GH:omdurman- rules/src/effects/ dispatch.rs:202	end_player_turn	<pre>200 201 /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook \$4). 202 pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> { 203 debug!(204 old_player = ?state.active_player,</pre>
omdurman- rules/src/ lib.rs:63 GH:omdurman- rules/src/lib.rs:63	GameTurnIn- dex::value	<pre>61 pub const ALL: &'static [Self] = &[\$(Self::: \$variant,)+]; 62 63 pub fn value(self) -> u16 { 64 match self { 65 \$(Self::\$variant => \$value,)+</pre>
omdurman- rules/src/effects/ state.rs:133 GH:omdurman- rules/src/effects/ state.rs:133	PendingMelee	<pre>131 /// resolution after the reaction window is deterministic and host-ordered (rulebook \$7.5). 132 #[derive(Serialize, Deserialize, Clone, Debug)] 133 pub struct PendingMelee { 134 pub attack: MeleeAttack, 135 pub attacker_roll: DieRoll,</pre>

Covered by tests: omdurman-rules::src::effects::tests::both_ready_auto_advances_out_of_setup
omdurman-rules::src::effects::tests::fire_combat_wrong_phase_rejected omdurman-rules::src::effects::tests::new_game_starts_in_setup
omdurman-rules::src::turn_track::scenario_turn_dispatches_correctly
omdurman-rules::src::effects::tests::turn_advances_through_phases
omdurman-rules::src::unit_profiles::game_turn_marker_cell_returns_none

18/23 implemented

§5 – Movement Phase

§5 – Movement Phase (general)

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Movement Phase

Covered by tests: `omdurman-rules::src::effects::tests::disrupted_unit_cannot_fire`

§5.1 – General Rules

descriptive

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General Rules

§5.2 – Movement Restrictions

descriptive

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Movement Restrictions

§5.3 – Constructing the Zariba

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Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:778 GH:omdurman-rules/src/lib.rs:778	constructing_zariba	<pre>776 /// Set while the unit is building Zariba hexsides -- -- neither offensive 777 /// fire nor melee allowed that turn (§5.3). 778 pub constructing_zariba: bool, 779 /// Set when the Royal Engineers are committed to a demolition this turn 780 /// (§6.53) -- neither offensive fire nor melee allowed that turn.</pre>
omdurman-rules/src/effects/effect.rs:169	ConstructZariba	<pre>167 168 /// Begin constructing a Zariba hexside (rulebook §5.3).</pre>

GH:omdurman-rules/src/effects/effect.rs:169		<pre> 169 ConstructZariba { 170 unit_ids: Vec<UnitId>, 171 hexside: HexsideRef, </pre>
omdurman-rules/src/effects/dispatch.rs:40 GH:omdurman-rules/src/effects/dispatch.rs:40	apply_construct_zariba	<pre> 38 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id), 39 GameEffect::ConstructZariba { unit_ids, hexside } => { 40 apply_construct_zariba(state, unit_ids, *hexside) 41 } 42 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target), </pre>

Covered by tests: [omdurman-rules::src::effects::tests::construct_zariba_marks_builders_and_records_hexside](#)

§5.4 – Zones of Control

[descriptive](#)

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Zones of Control

§5.5 – Stacking

[descriptive](#)

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Stacking

§5.11 – Movement allowances printed on units

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The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:131 GH:omdurman-rules/src/lib.rs:131	MovementAllowance	<pre> 129 /// is a named variant. 130 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 131 pub enum MovementAllowance { 132 /// Immobile (forts, wrecked gunboats). 133 Immobile = 0, </pre>
omdurman-rules/src/lib.rs:755 GH:omdurman-rules/src/lib.rs:755	UnitMovement	<pre> 753 /// Movement allowance -- uniform for land units, split for gunboats (rulebook §5.11, §5.24, §5.25). 754 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 755 pub enum UnitMovement { 756 Land(MovementAllowance), 757 Gunboat(GunboatMovement), </pre>
omdurman-types/src/lib.rs:302 GH:omdurman-types/src/lib.rs:302	HexDirection	<pre> 300 /// (`+q`, `+q+r`, `+r`, `~q`, `~q-r`, `~r` for pointy- top hexes) (rulebook §5.11, §5.24). 301 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, Default)] 302 pub enum HexDirection { 303 #[default] 304 East = 0, </pre>

omdurman-rules/src/lib.rs:173 GH:omdurman-rules/src/lib.rs:173	MovementPoints	<pre> 171 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 172 PartialOrd, Ord, Hash, Debug, Default, 173 pub struct MovementPoints(i16); 174 175 impl MovementPoints { </pre>
omdurman-rules/src/terrain_chart.rs:21 GH:omdurman-rules/src/terrain_chart.rs:21	terrain_effects_chart	<pre> 19 /// 20 /// Source: printed Terrain Effects Chart on the mapsheet. 21 pub fn terrain_effects_chart(terrain: Terrain) -> TerrainEntry { 22 match terrain { 23 Terrain::Clear { .. } => TerrainEntry { </pre>
omdurman-rules/src/terrain_chart.rs:77 GH:omdurman-rules/src/terrain_chart.rs:77	movement_cost	<pre> 75 /// Convenience: get the movement cost for a terrain type (rulebook §5.11, Terrain Effects Chart). 76 /// Returns `None` for impassable terrain (Nile). 77 pub fn movement_cost(terrain: Terrain) -> Option<MovementAllowance> { 78 terrain_effects_chart(terrain).movement_cost 79 } </pre>
omdurman-rules/src/terrain_chart.rs:85 GH:omdurman-rules/src/terrain_chart.rs:85	movement_cost_with_road	<pre> 83 /// underlying terrain; without a road it's the terrain's own cost. The road is 84 /// a movement overlay only -- combat/LOS still use the underlying terrain. 85 pub fn movement_cost_with_road(terrain: Terrain, road: bool) -> Option<MovementAllowance> { 86 if road { 87 Some(MovementAllowance::One) </pre>
omdurman-types/src/lib.rs:498 GH:omdurman-types/src/lib.rs:498	Terrain::has_road	<pre> 496 497 /// Whether this hex has any road touching it. 498 pub fn has_road(self) -> bool { 499 !matches!(self.road(), Road::None) 500 } </pre>
omdurman-types/src/lib.rs:435 GH:omdurman-types/src/lib.rs:435	Terrain::passable_by_land	<pre> 433 434 /// Whether this terrain may be entered by land units (rulebook §5.11). 435 pub fn passable_by_land(self) -> bool { 436 !self.is_nile() 437 } </pre>
omdurman-types/src/lib.rs:503 GH:omdurman-types/src/lib.rs:503	Terrain::is_crossroad	<pre> 501 502 /// Whether roads converge at this hex's centre. 503 pub fn is_crossroad(self) -> bool { 504 matches!(self.road(), Road::Crossroad) 505 } </pre>
omdurman-hexmap/src/map.rs:17 GH:omdurman-hexmap/src/map.rs:17	GameMap::roads	<pre> 15 pub hexes: HashMap<HexCoord, HexData>, 16 pub hexsides: HashMap<HexsideRef, HexsideKind>, 17 pub roads: HashSet<HexsideRef>, 18 pub excluded: HashSet<HexCoord>, 19 pub overlay: OverlayParams, </pre>

Covered by tests: `omdurman-rules::src::terrain_chart::clear_terrain_no_bonus` `omdurman-rules::src::terrain_chart::nile_is_impassable`
`omdurman-rules::src::terrain_chart::rough_movement_and_defense`
`omdurman-rules::src::terrain_chart::swamp_movement_and_defense`
`omdurman-rules::src::terrain_chart::hilltop_movement_and_defense`
`omdurman-rules::src::terrain_chart::huts_movement_and_defense`
`omdurman-rules::src::terrain_chart::movement_cost_convenience_matches_chart`
`omdurman-rules::src::terrain_chart::movement_cost_with_road_always_one`
`omdurman-rules::src::effects::tests::land_unit_may_not_enter_nile`
`omdurman-rules::src::terrain_chart::movement_cost_without_road_matches_terrain`
`omdurman-rules::src::effects::tests::movement_cost_for_uses_terrain`
`omdurman-rules::src::effects::tests::movement_cost_for_road_costs_one`

omdurman-rules::src::terrain_chart::road_gives_crossroad

omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds

omdurman-rules::src::terrain_chart::terrain_chart_road_always_costs_one

§5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

manual page unknown

A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:665 GH:omdurman-rules/src/effects/state.rs:665	can_move_unit	<pre>663 /// the same `RuleError` the `MoveUnit` effect 664 /// would on rejection. Lets the 665 /// UI gate input without mutating or duplicating 666 /// the rules. 667 pub fn can_move_unit(&self, unit_id: UnitId, cost: MovementPoints) -> Result<(), RuleError> { self.can_move_unit_to(unit_id, None, cost) }</pre>
omdurman-rules/src/effects/state.rs:1356 GH:omdurman-rules/src/effects/state.rs:1356	mp_spent	<pre>1354 1355 /// Movement points `unit_id` has already spent 1356 /// this turn (§5.11/§5.12). 1357 pub fn mp_spent(&self, unit_id: UnitId) -> i16 { 1358 self.mp_spent_this_turn.get(&unit_id).copied().unwrap_or(0) 1359 }</pre>

Covered by tests: omdurman-rules::src::effects::tests::cumulative_move_cost_may_not_exceed_allowance

§5.13 – No MP accumulation between turns

implemented

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A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:202 GH:omdurman-rules/src/effects/dispatch.rs:202	end_player_turn	<pre>200 201 /// End the current player's turn: recover disrupted 202 /// units, switch active player, advance turn index (rulebook 203 /// §4). 204 pub fn end_player_turn(state: &mut GameState) -> Result<(), RuleError> { debug!(old_player = ?state.active_player,</pre>

Covered by tests: omdurman-rules::src::effects::tests::unused_movement_points_do_not_carry_over

§5.21 – Friendlies transport via gunboat

implemented

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In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The

transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:592 GH:omdurman-rules/src/lib.rs:592	is_friendlylies	<pre> 590 /// "Friendlies" units obey several special rules (\$5.21, \$5.23, \$6.52, 591 /// \$9.14 victory conditions). pub fn is_friendlylies(&self) -> bool { 592 matches!(593 self, 594 self, </pre>
omdurman-rules/src/effects/state.rs:2053 GH:omdurman-rules/src/effects/state.rs:2053	friendlylies_transport_offer	<pre> 2051 /// regardless of selection. Pairs with [`GameEffect::FriendlyliesTransport`] 2052 /// so the UI can offer exactly the action the engine would accept. 2053 pub fn friendlylies_transport_offer(&self, selected: Option<UnitId>) -> Option<FriendlyliesAction> { 2054 match self.friendlylies_transport { 2055 None => { </pre>
omdurman-rules/src/lib.rs:775 GH:omdurman-rules/src/lib.rs:775	loaded_on	<pre> 773 pub disrupted: bool, 774 /// `Some(gunboat)` after a "Friendlies" unit loads onto a gunboat (\$5.21). 775 pub loaded_on: Option<UnitId>, 776 /// Set while the unit is building Zariba hexsides -- neither offensive 777 /// fire nor melee allowed that turn (\$5.3). </pre>
omdurman-rules/src/effects/effect.rs:196 GH:omdurman-rules/src/effects/effect.rs:196	FriendlyliesTransport	<pre> 194 195 /// Load/disembark the "Friendlies" brigade via gunboat (rulebook \$5.21). 196 FriendlyliesTransport(crate::FriendlyliesAction), 197 198 // -- Optional rules ----- </pre>
omdurman-rules/src/effects/river.rs:12 GH:omdurman-rules/src/effects/river.rs:12	apply_friendlylies_transport	<pre> 10 /// unit is freed (a disembarking `MoveUnit` should follow, costed by 11 /// terrain). 12 pub fn apply_friendlylies_transport(13 state: &mut GameState, 14 action: FriendlyliesAction, </pre>
omdurman-rules/src/lib.rs:1013 GH:omdurman-rules/src/lib.rs:1013	FriendlyliesAction	<pre> 1011 /// tracks each unit–gunboat pair independently. 1012 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1013 pub enum FriendlyliesAction { 1014 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1015 /// loads onto (stacks with) the gunboat. </pre>
omdurman-rules/src/lib.rs:1033 GH:omdurman-rules/src/lib.rs:1033	TransportState	<pre> 1031 /// third turn. 1032 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1033 pub enum TransportState { 1034 /// Turn N (the load turn): unit and gunboat started adjacent; unit 1035 /// loads onto (stacks with) the gunboat. </pre>

Covered by tests: [omdurman-rules::src::effects::tests::friendlylies_transport_offer_load_requires_prerequisites](#)
[omdurman-rules::src::effects::tests::friendlylies_transport_offer_follows_state_machine](#)

§5.22 – Land units may never enter a Nile River hex

implemented

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With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:682 GH:omdurman-rules/src/effects/state.rs:682	can_move_unit_to	<pre>680 /// 681 /// [<code>`hex_in_enemy_zoc`</code>]: Self::hex_in_enemy_zoc 682 pub fn can_move_unit_to(683 &self, 684 unit_id: UnitId,</pre>
omdurman-rules/src/effects/state.rs:353 GH:omdurman-rules/src/effects/state.rs:353	in_deployment_zone	<pre>351 /// plan / UI rather than this hex predicate. 352 Documented, not silently 353 /// dropped. 354 pub fn in_deployment_zone(&self, player: Player, 355 hex: HexCoord, is_boat: bool) -> bool { 356 // No board attached -> permissive (unit tests, 357 unbound session). 358 if self.board.terrain.is_empty() {</pre>

Covered by tests: [omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile](#)
[omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine](#)
[omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile](#)
[omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_nile](#)

§5.23 – Walled city entry restrictions

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Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:152 GH:omdurman-types/src/lib.rs:152	HexsideRef	<pre>150 /// data by [<code>`HexsideRef`</code>]. 151 #[derive(Serialize, Deserialize, Clone, Copy, 152 PartialEq, Eq, Hash, Debug)] 153 pub struct HexsideRef { 154 pub a: HexCoord,</pre>
omdurman-types/src/lib.rs:187 GH:omdurman-types/src/lib.rs:187	HexsideKind	<pre>185 strum::EnumIter, 186)] 187 pub enum HexsideKind { 188 /// City wall (Khartoum, walled city of Omdurman). 189 Blocks LOS, blocks 190 /// movement except across gates/breaches (§5.23), 191 blocks ZOC into the city</pre>
omdurman-types/src/lib.rs:258	blocks_movement	<pre>256 /// `omdurman-rules`. The trench *end* variants 257 are therefore intentionally 258 /// not blocking. 259 pub fn blocks_movement(self) -> bool {</pre>

GH:omdurman-types/src/lib.rs:258		<pre> 259 matches!(260 self, </pre>
omdurman-rules/src/lib.rs:618 GH:omdurman-rules/src/lib.rs:618	may_enter_walled_city	<pre> 616 /// Taiasha bodyguard may enter. Anglo-Egyptian: any unit that can reach the 617 /// walled city *except* gunboats and "Friendlies". 618 pub fn may_enter_walled_city(&self) -> bool { 619 match self { 620 // \$5.23 Dervish: Khalifa, artillery, Taiasha. </pre>
omdurman-rules/src/board.rs:294 GH:omdurman-rules/src/board.rs:294	is_walled_city	<pre> 292 /// are its seeds). The set is derived once from the board data, replacing 293 /// the older "at least two of six hexsides are Wall/Gate/Breach" heuristic. 294 pub fn is_walled_city(&self, hex: HexCoord) -> bool { 295 // Membership in the precomputed enclosed area (see `walled_city`). 296 // Palace/Tomb hexes are always part of it (they are the seeds). </pre>
omdurman-rules/src/effects/error.rs:90 GH:omdurman-rules/src/effects/error.rs:90	WalledCityEntry	<pre> 88 89 #[error("unit {0:?} is not eligible to enter the walled city of Omdurman at {1:?} (\$5.23)")] 90 WalledCityEntry(UnitId, HexCoord), 91 92 #[error("movement cost {cost:?} exceeds allowance {allowance:?}")] </pre>

Proven by: [omdurman-types::src::lib::hexside_ref_is_order_independent](#)

Covered by tests:

[omdurman-rules::src::effects::tests::can_move_rejects_wall_hexside](#)
[omdurman-rules::src::effects::tests::can_move_allows_gate_hexside](#)
[omdurman-rules::src::effects::tests::walled_city_entry_allows_khalifa](#)
[omdurman-rules::src::effects::tests::walled_city_entry_rejects_unauthorized_dervish](#)
[omdurman-rules::src::effects::tests::walled_city_entry_rejects_ae_gunboat](#)
[omdurman-rules::src::effects::tests::walled_city_entry_not_enforced_for_fok](#)
[omdurman-rules::src::board_data::campaign_walled_city_is_enclosed_by_walls](#)

§5.24 – Gunboat upstream/downstream movement

implemented

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Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:528 GH:omdurman-rules/src/lib.rs:528	GunboatMovement	<pre> 526 /// the turn. 527 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 528 pub struct GunboatMovement { </pre>

		529 <code>pub upstream: MovementAllowance,</code> 530 <code>pub downstream: MovementAllowance,</code>
omdurman-types/src/lib.rs:949 GH:omdurman-types/src/lib.rs:949	is_boat	947 948 <code>/// Gunboats use the split upstream/downstream</code> <code>movement allowance (§5.24).</code> 949 <code>pub fn is_boat(self) -> bool {</code> 950 <code>matches!(self, UnitKind::Gunboat { .. })</code> 951 <code>}</code>

Covered by tests: `omdurman-rules::src::unit_profiles::boat_annotation_yields_split_gunboat_movement`
`omdurman-rules::src::effects::tests::gunboat_upstream_cap_is_sticky_across_moves`

§5.25 – Dervish forts may not move

implemented

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Dervish forts may not move in any way once placed.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:759 GH:omdurman-rules/src/lib.rs:759	Immobile	757 <code>Gunboat(GunboatMovement),</code> 758 <code>/// Forts may not move once placed (§5.25).</code> 759 <code>Immobile,</code> 760 <code>}</code> 761
omdurman-types/src/lib.rs:906 GH:omdurman-types/src/lib.rs:906	UnitKind::Fort	904 <code>/// Permanent emplacement (§6.54): fire</code> <code>(artillery) / melee (defensive).</code> 905 <code>/// May not move once placed (§5.25).</code> 906 <code>Fort { fire: i32, melee: i32 },</code> 907 <code>/// Dervish leader (§6.51): fire / melee /</code> <code>movement. May melee attack (§7.4).</code> 908 <code>DervishLeader {</code>
omdurman-rules/src/lib.rs:759 GH:omdurman-rules/src/lib.rs:759	UnitMovement::Immobile	757 <code>Gunboat(GunboatMovement),</code> 758 <code>/// Forts may not move once placed (§5.25).</code> 759 <code>Immobile,</code> 760 <code>}</code> 761

Covered by tests: `omdurman-rules::src::effects::tests::forts_are_never_advance_eligible`

§5.26 – Units stop on entering enemy ZOC

implemented

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Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

See also: §5.4

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:682 GH:omdurman-rules/src/effects/state.rs:682	can_move_unit_to	680 <code>///</code> 681 <code>/// [<code>hex_in_enemy_zoc</code>]: Self::hex_in_enemy_zoc</code> 682 <code>pub fn can_move_unit_to{</code> 683 <code>&self,</code> 684 <code>unit_id: UnitId,</code>
omdurman-rules/src/effects/	hex_in_enemy_zoc	1617 <code>/// does not extend into or out of a Nile hex.</code> <code>With no board loaded these</code>

state.rs:1619 GH:omdurman-rules/src/effects/state.rs:1619	<pre> 1618 /// reduce to the plain adjacency rule. 1619 pub fn hex_in_enemy_zoc(1620 &self, 1621 hex: HexCoord, </pre>
--	---

Covered by tests: [omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn](#)
[omdurman-rules::src::effects::tests::zoc_transit_check_uses_the_actual_path](#)

§5.41 – All units except AE leaders exert ZOC

implemented

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All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:814 GH:omdurman-rules/src/lib.rs:814	ZocReason	<pre> 812 /// Used by the engine when answering "is this hex in an enemy ZOC?". 813 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 814 pub enum ZocReason { 815 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 816 /// leader (§5.41) projects ZOC into each of its six adjacent hexes. </pre>
omdurman-rules/src/effects/state.rs:1585 GH:omdurman-rules/src/effects/state.rs:1585	unit_projects_zoc	<pre> 1583 /// §5.44) need the game map, which the engine does not hold; the app layers 1584 /// those on top. This is the position/kind/ disruption core of the rule. 1585 pub fn unit_projects_zoc(1586 &self, 1587 unit: &UnitPlacement, </pre>
omdurman-rules/src/effects/state.rs:1660 GH:omdurman-rules/src/effects/state.rs:1660	zoc_hexes	<pre> 1658 /// ZOC covers a given hex; this function returns *which* hexes a 1659 /// specific unit covers. 1660 pub fn zoc_hexes(1661 &self, 1662 unit: &UnitPlacement, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::zoc_hexes_empty_for_anglo_egyptian_leader](#)
[omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions](#)
[omdurman-rules::src::effects::tests::zoc_hexes_empty_for_disrupted_unit](#)
[omdurman-rules::src::effects::tests::zoc_hexes_matches_hex_in_enemy_zoc](#) [omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile](#)
[omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor](#)

§5.42 – No MP cost to enter/leave enemy ZOC

implemented

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There is no movement point cost to enter or leave an enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:848 GH:omdurman-rules/src/effects/state.rs:848	movement_cost_for	<pre> 846 /// 847 /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 848 pub fn movement_cost_for(849 &self, 850 unit: &UnitPlacement, </pre>

Covered by tests: `omdurman-rules::src::effects::tests::entering_enemy_zoc_costs_no_extra_mp`

§5.43 – Units stop when entering enemy ZOC

implemented

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All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:682 GH:omdurman-rules/src/effects/state.rs:682	can_move_unit_to	<pre>680 /// 681 /// [<code>hex_in_enemy_zoc</code>]: Self::<code>hex_in_enemy_zoc</code> 682 pub fn can_move_unit_to(683 &self, 684 unit_id: UnitId,</pre>
omdurman-rules/src/effects/state.rs:1619 GH:omdurman-rules/src/effects/state.rs:1619	hex_in_enemy_zoc	<pre>1617 /// does not extend into or out of a Nile hex. With no board loaded these 1618 /// reduce to the plain adjacency rule. 1619 pub fn hex_in_enemy_zoc(1620 &self, 1621 hex: HexCoord,</pre>

Covered by tests: `omdurman-rules::src::effects::tests::unit_entering_enemy_zoc_may_move_no_further_that_turn`

§5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

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ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:814 GH:omdurman-rules/src/lib.rs:814	ZocReason	<pre>812 /// Used by the engine when answering "is this hex in an enemy ZOC?". 813 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 814 pub enum ZocReason { 815 /// Normal ZOC: any non-disrupted unit other than an Anglo-Egyptian 816 /// leader (§5.41) projects ZOC into each of its six adjacent hexes.</pre>
omdurman-types/src/lib.rs:192	Wall	<pre>190 /// (§5.44), blocks melee (§7.2), blocks advance- after-combat (§6.82). 191 #[default] 192 Wall,</pre>

GH:omdurman-types/src/lib.rs:192		<pre> 193 /// Gate hexside in a wall. ZOC extends *out of* the walled city through 194 /// gates but not into it (\$5.44). Melee may be made through a gate (\$7.2). </pre>
omdurman-types/src/lib.rs:201 GH:omdurman-types/src/lib.rs:201	Khor	<pre> 199 /// Khor -- gully/wadi. ZOCs do not extend across (\$5.44); advance after 200 /// combat may not cross (\$6.82). 201 Khor, 202 /// Crest line. Blocks LOS unless the firer is on the higher side 203 /// (\$6.3 note 7). </pre>
omdurman-rules/src/lib.rs:827 GH:omdurman-rules/src/lib.rs:827	ZocReason::Zariba	<pre> 825 /// Zariba hexside ZOC behaviour in the historical scenario / when the 826 /// Zariba is constructed (\$5.44). 827 Zariba, 828 } 829 </pre>
omdurman-rules/src/effects/state.rs:1585 GH:omdurman-rules/src/effects/state.rs:1585	unit_projects_zoc	<pre> 1583 /// \$5.44) need the game map, which the engine does not hold; the app layers 1584 /// those on top. This is the position/kind/ disruption core of the rule. 1585 pub fn unit_projects_zoc(1586 &self, 1587 unit: &UnitPlacement, </pre>
omdurman-types/src/lib.rs:275 GH:omdurman-types/src/lib.rs:275	HexsideKind::blocks_zoc	<pre> 273 /// cannot express; those are left to the caller. This predicate captures the 274 /// symmetric "does not extend across" cases. 275 pub fn blocks_zoc(self) -> bool { 276 matches!(277 self, </pre>
omdurman-rules/src/effects/state.rs:1619 GH:omdurman-rules/src/effects/state.rs:1619	hex_in_enemy_zoc	<pre> 1617 /// does not extend into or out of a Nile hex. With no board loaded these 1618 /// reduce to the plain adjacency rule. 1619 pub fn hex_in_enemy_zoc(1620 &self, 1621 hex: HexCoord, </pre>

Covered

by tests: `omdurman-rules::src::effects::tests::zoc_hexes_excludes_nile` `omdurman-rules::src::effects::tests::zoc_hexes_excludes_khor` `omdurman-rules::src::effects::tests::zoc_hexes_normal_unit_projects_six_adjacent_minus_exclusions`

§5.51 – Stacking limit (4 units + leaders, gunboats isolated)

implemented

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No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:836 GH:omdurman-rules/src/lib.rs:836	OverLimit	<pre> 834 /// and the gunboat exception. 835 #[error("hex stack exceeds the four-unit limit")] 836 OverLimit, 837 /// "Gunboats may not stack with any other unit" (\$5.51, exception §5.21). 838 #[error("gunboats may not stack with non-gunboat units")] </pre>

omdurman-rules/src/lib.rs:839 GH:omdurman-rules/src/lib.rs:839	GunboatStack	<pre> 837 /// "Gunboats may not stack with any other unit" (\$5.51, exception \$5.21). 838 #[error("gunboats may not stack with non-gunboat units")] GunboatStack, 840 /// "Units of different Dervish tribes may not stack together" (\$5.52). #[error("Dervish units of different tribes may not stack")] </pre>
omdurman-rules/src/effects/state.rs:1412 GH:omdurman-rules/src/effects/state.rs:1412	check_stacking	<pre> 1410 /// * \$5.52 -- units of different Dervish tribes may not stack together. /// * \$5.53 -- a Dervish leader may stack only with units of its command. 1412 pub fn check_stacking(&self, 1413 mover: &UnitPlacement, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::stacking_over_limit_rejected](#)
[omdurman-rules::src::effects::tests::mid_move_stacking_allows_pass_through](#)
[omdurman-rules::src::effects::tests::mid_move_stacking_rejects_over_limit_destination](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_catches_stacking_violation](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_allows_leaders_stacking](#)

§5.52 – Different Dervish tribes may not stack together

implemented

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The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:842 GH:omdurman-rules/src/lib.rs:842	DervishTribeMix	<pre> 840 /// "Units of different Dervish tribes may not stack together" (\$5.52). #[error("Dervish units of different tribes may not stack")] DervishTribeMix, 843 /// "If Dervish leaders elect to stack, they may only stack with units of their command (i.e. colour)" (\$5.53). 844 </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::green_sections_are_mulazmin_tribal_units](#)
[omdurman-rules::src::effects::tests::deploy_rejects_dervish_tribe_mix](#)
[omdurman-rules::src::effects::tests::validate_stacking_invariants_clean_state](#)

§5.53 – Leader stacking with command colour only

implemented

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Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

****5.54) Anglo-Egyptian Brigade Integrity:**** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

See also: §5.54, §6.24

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:846 GH:omdurman-rules/src/lib.rs:846	DervishLeaderCommandMismatch	<pre> 844 /// their command (i.e. colour)" (§5.53). 845 #[error("Dervish leader may only stack with units of their own command")] 846 DervishLeaderCommandMismatch, 847 } 848 </pre>

Covered by tests: `omdurman-rules::src::effects::tests::dervish_leader_stacks_only_with_command_colour`

§5.54 – Anglo-Egyptian Brigade Integrity

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:852 GH:omdurman-rules/src/lib.rs:852	BrigadeIntegrity	<pre> 850 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 851 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 852 pub enum BrigadeIntegrity { 853 None, 854 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:717 GH:omdurman-rules/src/lib.rs:717	brigade_integrity	<pre> 715 /// Only a full stack of four battalions qualifies. Three or fewer may still 716 /// stack and fire, but they receive no brigade- integrity bonus. 717 pub fn brigade_integrity(identities: &[UnitIdentity]) - > BrigadeIntegrity { 718 let Some(brigade) = identities.first().and_then(i i.brigade()) else { 719 return BrigadeIntegrity::None; </pre>
omdurman-rules/src/lib.rs:852 GH:omdurman-rules/src/lib.rs:852	FireModifier::BrigadeIntegrity	<pre> 850 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 851 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 852 pub enum BrigadeIntegrity { 853 None, 854 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:329 GH:omdurman-rules/src/lib.rs:329	BattalionOrdinal	<pre> 327 /// brigade integrity requires all four stacked in one hex (§5.54). 328 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 329 pub enum BattalionOrdinal { 330 First = 1, 331 Second = 2, </pre>
omdurman-types/src/lib.rs:1014 GH:omdurman-types/src/lib.rs:1014	BrigadeId	<pre> 1012 /// same field for uniform handling. 1013 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 1014 pub struct BrigadeId { 1015 pub number: u8, 1016 pub nationality: BrigadeNationality, </pre>

Covered by tests: `omdurman-rules::src::unit_profiles::brigade_designation_ignored_for_non_infantry`
`omdurman-rules::src::unit_profiles::printed_brigade_designation_overrides_column`
`omdurman-rules::src::unit_profiles::tribe_stats_come_from_annotation`
`omdurman-rules::src::unit_profiles::section_owner_anglo_egyptian_sections`
`omdurman-rules::src::unit_profiles::section_owner_green_sections_are_dervish`

omdurman-rules::src::lib::brigade_integrity_four_battalions_returns_integrated
omdurman-rules::src::unit_profiles::ae_infantry_fourth_battalion_from_col_3
omdurman-rules::src::lib::brigade_integrity_empty_slice omdurman-rules::src::lib::brigade_integrity_friendlylies_returns_none
omdurman-rules::src::unit_profiles::section_owner_dervish_sections
omdurman-rules::src::lib::brigade_integrity_three_battalions_returns_none
omdurman-rules::src::lib::unit_identity_brigade_and_battalion_accessors
omdurman-rules::src::unit_profiles::ae_infantry_brigade_number_three_from_col_7
omdurman-rules::src::lib::brigade_integrity_non_infantry_returns_none
omdurman-rules::src::lib::brigade_integrity_mixed_brigades_returns_none
omdurman-rules::src::unit_profiles::ae_infantry_third_battalion_from_col_2
omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived

21/31 implemented

§6 – Fire Combat Phase

§6 – Fire Combat Phase

descriptive

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Fire Combat Phase

§6.1 – General Rules

descriptive

manual page unknown

General Rules

§6.2 – How To Have Fire Combat

descriptive

manual page unknown

How To Have Fire Combat

§6.3 – Line of Sight Table

implemented

manual page unknown

Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:57 GH:omdurman-rules/src/los_table.rs:57	LosLevel	<pre>55 serde::Serialize, serde::Deserialize, Clone, Copy, 56 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 57)] 58 pub enum LosLevel { 59 Ground, 59 Rough,</pre>
omdurman-rules/src/los_table.rs:69 GH:omdurman-rules/src/los_table.rs:69	LosFeature	<pre>67 /// authored RON spellings. 68 #[derive(serde::Serialize, serde::Deserialize, Clone, 69 Copy, PartialEq, Eq, Hash, Debug)] 70 pub enum LosFeature { 71 /// A hex containing units (gunboats/forts excluded 71 Units,</pre>
omdurman-rules/src/los_table.rs:91 GH:omdurman-rules/src/los_table.rs:91	LosCondition	<pre>89 /// A positional condition from the LOS table Detail 90 footnotes. 91 #[derive(serde::Serialize, serde::Deserialize, Clone, 92 Copy, PartialEq, Eq, Hash, Debug)] 93 pub enum LosCondition { 94 /// (1) Blocks only if the ray passes through more</pre>

		than two such features. 93 MoreThanTwo,
omdurman-rules/src/los_table.rs:170 GH:omdurman-rules/src/los_table.rs:170	los_level	168 /// 169 /// For all other units, the level is derived from the terrain at `hex`. 170 pub fn los_level_for_unit(171 kind: UnitKind, 172 hex: HexCoord,
omdurman-rules/src/los_table.rs:170 GH:omdurman-rules/src/los_table.rs:170	los_level_for_unit	168 /// 169 /// For all other units, the level is derived from the terrain at `hex`. 170 pub fn los_level_for_unit(171 kind: UnitKind, 172 hex: HexCoord,
omdurman-rules/src/los_table.rs:209 GH:omdurman-rules/src/los_table.rs:209	blocking_rules	207 /// of its conditions are satisfied (AND semantics); an empty conditions list 208 /// means the feature always blocks. 209 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 210 let table = crate::tables_data::los_table_data(); 211 match table.cells.get(&(firer, target)) {
omdurman-rules/src/los_table.rs:295 GH:omdurman-rules/src/los_table.rs:295	has_los	293 /// `unit_level_at` closure returns the LOS level of blocking units 294 /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`. 295 pub fn has_los(296 board: &crate::board::BoardInfo, 297 from: HexCoord,
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	HexsideKind::blocks_los	228 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 229 /// in `omdurman_rules::los_table`, not by this predicate. 230 pub fn blocks_los(self) -> bool { 231 matches!(self, HexsideKind::Wall HexsideKind::Crest) 232 }
omdurman-types/src/lib.rs:230 GH:omdurman-types/src/lib.rs:230	Terrain::blocks_los	228 /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine 229 /// in `omdurman_rules::los_table`, not by this predicate. 230 pub fn blocks_los(self) -> bool { 231 matches!(self, HexsideKind::Wall HexsideKind::Crest) 232 }
omdurman-types/src/lib.rs:452 GH:omdurman-types/src/lib.rs:452	Terrain::is_los_trees	450 /// (\$6.3 note 1). Retained for compatibility; the full LOS engine 451 /// checks `Terrain::Trees` directly. 452 pub fn is_los_trees(self) -> bool { 453 matches!(self, Terrain::Trees { .. }) 454 }

Proven by: [omdurman-types::src::lib::line_between_forms_a_connected_ray](#)

Covered by tests: [omdurman-rules::src::los_table::los_level_mapping](#) [omdurman-rules::src::los_table::blocking_rules_all_cells_covered](#)
[omdurman-rules::src::los_table::has_los_empty_board_is_clear](#) [omdurman-rules::src::los_table::has_los_adjacent_clear](#)
[omdurman-rules::src::los_table::has_los_howitzzer_bypasses](#)
[omdurman-rules::src::los_table::has_los_wall_hexside_blocks_ground_to_ground](#)
[omdurman-rules::src::los_table::has_los_gate_hexside_passes](#) [omdurman-rules::src::los_table::has_los_breach_hexside_passes](#)
[omdurman-rules::src::los_table::has_los_rough_intervening_blocks_ground_to_ground](#)
[omdurman-rules::src::los_table::has_los_two_tree_hexes_pass_ground_to_ground](#)
[omdurman-rules::src::los_table::has_los_three_tree_hexes_block_ground_to_ground](#)
[omdurman-rules::src::los_table::has_los_two_hut_hexes_pass_ground_to_ground](#)
[omdurman-rules::src::los_table::has_los_three_hut_hexes_block_ground_to_ground](#)

```

omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_clear_no_units
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_blocked_by_hilltop_unit
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_lower_level_passes
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_same_level_blocks
omdurman-rules::src::los_table::has_los_rough_to_rough_hilltop_blocks
omdurman-rules::src::los_table::has_los_ground_to_hilltop_intervening_hilltop_blocks
omdurman-rules::src::los_table::has_los_building_blocks_like_huts_ground_to_ground
omdurman-rules::src::los_table::has_los_two_building_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::los_level_for_unit_gunboat_is_rough
omdurman-rules::src::los_table::los_level_for_unit_fort_is_ground
omdurman-rules::src::los_table::los_level_for_unit_walled_city_adj_wall_is_rough
omdurman-rules::src::los_table::gunboat_firer_uses_rough_row_not_ground
omdurman-rules::src::los_table::los_reflexive_all_terrains
omdurman-rules::src::los_table::los_reflexive_hilltop
omdurman-rules::src::los_table::los_symmetric_ground_to_ground_no_units
omdurman-rules::src::los_table::los_howitzer_always_has_los
omdurman-rules::src::los_table::los_howitzer_same_hex
omdurman-rules::src::los_table::los_blocking_rules_match_reference_table
omdurman-rules::src::los_table::los_ground_to_ground_features_block_as_expected

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§6.4 – Fire Combat Sequence

implemented

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Fire Combat Sequence

The sequence of fire combat resolution is the same for both defensive and offensive fire combat. During the Dervish player turn the Anglo-Egyptian player executes 6.41 AND 6.42 as defensive fire, after which the Dervish player executes 6.41 as offensive fire. During the Anglo-Egyptian player turn the Dervish player executes 6.41 as defensive fire, after which the Anglo-Egyptian player executes 6.41 AND 6.42 as offensive fire.

****6.41) Direct Fire Subphase (Dervish and Anglo-Egyptian players):**** The firing player must first allocate all of his fire attacks, combining his units' direct fire combat factors in any manner he wishes. After all fire has been allocated, the firing player then resolves his attacks in any order he wishes.

****6.42) Maxim Second Fire and Howitzer Fire Subphase (Anglo-Egyptian player only):**** Anglo-Egyptian named gunboats may now fire their artillery factor as howitzer fire (see 6.64) and all Maxim guns may fire a second time. Once

(see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:302 GH:omdurman-rules/src/lib.rs:302	FireSubPhase	<pre> 300 301 #[derive(Serialize, Deserialize, Clone, Copy, 302 PartialEq, Eq, Debug)] 303 pub enum FireSubPhase { 304 /// Direct fire (§6.41). Both sides participate in this sub-phase. 305 DirectFire, </pre>
omdurman-rules/src/effects/effect.rs:72 GH:omdurman-rules/src/effects/effect.rs:72	FireCombat	<pre> 70 /// - Firers marked as fired; target hex marked as fired-at. 71 /// - Victory points awarded for eliminations. 72 FireCombat { attack: FireAttack, roll: DieRoll }, 73 74 /// Resolve a howitzer bombardment (two rolls: CRT + impact scatter) </pre>

Covered by tests: [omdurman-rules::src::effects::tests::turn_advances_through_phases](#)

§6.5 – Special Unit Capabilities

descriptive

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Special Unit Capabilities

****6.51) Leader Units:**** Dervish leader units have a fire factor, a melee factor, and a movement factor. They may thus attack, melee, and be eliminated like any other combat unit. Their special benefit is that they stack free.

Anglo-Egyptian leaders have a movement factor only. They are eliminated if a) they are alone in a hex when a Dervish unit occupies or passes through that hex, or b) if all of the combat units a leader is stacked with are eliminated in fire combat or melee. The special function of Anglo-Egyptian leaders is that at least one must survive to occupy the Mahdi's tomb hex if it is to be taken from the Dervish player (see 9.14).

****6.52) Anglo-Egyptian "Friendlies" Brigade:**** These units represent native volunteers in the Anglo-Egyptian army. They fire rifles on the Dervish Range Effects Table and melee with the Dervish melee modifier. They may not enter the walled city of Omdurman (see 5.23). They may be transferred to the west bank (see 5.21).

(see manual for full text)

See also: §5.21, §5.23, §5.44, §6.51, §6.52, §6.53, §6.54, §6.62, §6.63, §7.6, §9.14

§6.6 – Special Artillery Capabilities

descriptive

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Special Artillery Capabilities

§6.7 – Defensive Fire

implemented

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Defensive Fire

In Defensive Fire phase, all of the non-moving player's units may fire at any of the moving player's units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.1, §6.6

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:1922 GH:omdurman-rules/src/effects/state.rs:1922	<code>can_advance_after_comb</code>	<pre>1920 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 1921 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 1922 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 1923 let unit = self.unit_or_err(unit_id)?; 1924 // §6.7: there is no advance after combat as a result of defensive fire.</pre>

Covered by tests: `omdurman-rules::src::effects::tests::no_advance_after_defensive_fire`
`omdurman-rules::src::effects::tests::defensive_fire_opens_no_advance_window`

§6.8 – Offensive Fire

descriptive

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Offensive Fire

§6.11 – Fire combat factor printed on units

implemented

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The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:86 GH:omdurman-rules/src/lib.rs:86	FireFactor	<pre>84 /// Every possible value from the annotated counter 85 set is a named variant. 86 #[derive(Serialize, Deserialize, Clone, Copy, 87 PartialEq, Eq, Hash, Debug, strum::Display)] 88 pub enum FireFactor { 89 One = 1, 90 Three = 3,</pre>

Covered by tests: [omdurman-rules::src::lib::fire_factor_sum_to_row](#)

§6.12 – Fire combat is always voluntary

implicit

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Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

§6.13 – Fire factor is unitary (may not be divided)

implicit

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If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

§6.14 – Players may combine fire factors into one attack

implemented

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Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

See also: §6.4

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:100 GH:omdurman-rules/src/lib.rs:100	sum_to_row	<pre>98 impl FireFactor { 99 /// Sum multiple fire factors and return the 100 corresponding Combat Results Table row (rulebook §6.11). 101 pub fn sum_to_row<'a>(factors: impl 102 IntoIterator<Item = &'a FireFactor>) -> FireFactorRow { 103 let total: u16 = factors.into_iter().map(f 104 f.value()).sum(); 105 crate::combat_results_table::FireFactorRow::from_total(total)</pre>

Covered by tests: [omdurman-rules::src::effects::tests::unit_may_only_be_fired_at_once_per_phase](#)

[omdurman-rules::src::effects::tests::gunboat_and_maxim_may_be_fired_at_repeatedly](#)

[omdurman-rules::src::effects::tests::rejected_fire_attack_does_not_mark_firers_as_fired](#)

§6.15 – May divide a stack to fire at different hexes

implicit

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Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

§6.16 – Halving fire strength rounds down, minimum 1

implemented

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When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:495 GH:omdurman-rules/src/lib.rs:495	RangeBand	<pre>493 /// multiplied at a given distance (§6.22). 494 #[derive(Serialize, Deserialize, Clone, Copy, 495 pub enum RangeBand { 496 Tripled, 497 Doubled,</pre>

Covered by tests: [omdurman-rules::src::lib::halving_rounds_down_and_never_below_one](#)

§6.21 – First check LOS before firing

implemented

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When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

File	Symbol	Code Snippet
omdurman-rules/src/los_table.rs:209 GH:omdurman-rules/src/los_table.rs:209	blocking_rules	<pre>207 /// of its conditions are satisfied (AND semantics); an empty conditions list 208 /// means the feature always blocks. 209 pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] { 210 let table = crate::tables_data::los_table_data(); 211 match table.cells.get(&(firer, target)) {</pre>
omdurman-rules/src/los_table.rs:295 GH:omdurman-rules/src/los_table.rs:295	has_los	<pre>293 /// `unit_level_at` closure returns the LOS level of blocking units 294 /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`. 295 pub fn has_los(296 board: &crate::board::BoardInfo, 297 from: HexCoord,</pre>

Covered

by

tests:

[omdurman-rules::src::effects::tests::can_fire_at_rejects_blocked_los](#) [omdurman-rules::src::effects::tests::can_fire_at_allows_clear_los](#)

§6.22 – Consult Range Effects Table

implemented

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Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

File	Symbol	Code Snippet
omdurman-rules/src/range_effects.rs:35 GH:omdurman-rules/src/range_effects.rs:35	ae_range_effects	<pre>33 /// Look up the range band for an Anglo-Egyptian weapon (\$6.22, §6.24). 34 /// Distances > 10 are out of range for all weapons. 35 pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 36 band_at(faction_rows(true), weapon, distance) 37 }</pre>
omdurman-rules/src/range_effects.rs:41 GH:omdurman-rules/src/range_effects.rs:41	dervish_range_effects	<pre>39 /// Look up the range band for a Dervish weapon (\$6.22). 40 /// Distances > 10 are out of range for all weapons. 41 pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand { 42 band_at(faction_rows(false), weapon, distance) 43 }</pre>
omdurman-rules/src/lib.rs:495 GH:omdurman-rules/src/lib.rs:495	RangeBand	<pre>493 /// multiplied at a given distance (§6.22). 494 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 495 pub enum RangeBand { 496 Tripled, 497 Doubled,</pre>
omdurman-rules/src/lib.rs:187 GH:omdurman-rules/src/lib.rs:187	HexDistance	<pre>185 /// (rulebook §6.22, §7.5). 186 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 187 pub struct HexDistance(u16); 188 189 impl HexDistance {</pre>

Proven by: [omdurman-types::src::lib::adjacency_iff_distance_one](#)

Covered by tests:

- [omdurman-rules::src::range_effects::ae_rifles_doubled_at_range_1](#)
- [omdurman-rules::src::range_effects::ae_rifles_halved_at_range_4](#)
- [omdurman-rules::src::range_effects::ae_howitzzer_range](#)
- [omdurman-rules::src::range_effects::dervish_rifles_shorter_range](#)
- [omdurman-rules::src::range_effects::melee_only_range_1](#)
- [omdurman-rules::src::range_effects::ae_range_effects_artillery_full](#)
- [omdurman-rules::src::range_effects::ae_range_effects_maxims_match_rifles](#)
- [omdurman-rules::src::range_effects::ae_range_effects_distance_over_10](#)
- [omdurman-rules::src::range_effects::ae_range_effects_howitzzer_halved_4_to_10](#)
- [omdurman-rules::src::range_effects::dervish_range_effects_rifles](#)
- [omdurman-rules::src::range_effects::dervish_range_effects_artillery](#)
- [omdurman-rules::src::range_effects::dervish_range_effects_maxims_and_howitzzer](#)
- [omdurman-rules::src::range_effects::dervish_range_effects_melee](#)
- [omdurman-rules::src::range_effects::dervish_range_effects_distance_over_10](#)
- [omdurman-rules::src::effects::tests::fire_combat_eliminate_target](#)
- [omdurman-rules::src::range_effects::max_day_range_all_combos](#)
- [omdurman-rules::src::range_effects::range_effects_every_cell_ae_artillery](#)
- [omdurman-rules::src::range_effects::range_effects_every_cell_ae_howitzzer](#)
- [omdurman-rules::src::range_effects::range_effects_every_cell_ae_maxims](#)
- [omdurman-rules::src::range_effects::range_effects_every_cell_ae_rifles](#)
- [omdurman-rules::src::range_effects::range_effects_every_cell_dervish_artillery](#)
- [omdurman-rules::src::range_effects::range_effects_every_cell_dervish_maxims_howitzzer](#)
- [omdurman-rules::src::range_effects::range_effects_every_cell_dervish_rifles](#)
- [omdurman-rules::src::range_effects::range_effects_every_cell_dervish_spears](#)
- [omdurman-rules::src::effects::tests::can_fire_at_gates_phase_range_and_player](#)
- [omdurman-rules::src::effects::tests::mixed_attack_bands_per_firer](#)
- [omdurman-rules::src::range_effects::ae_range_effects_monotone_non_increasing](#)
- [omdurman-rules::src::range_effects::ae_howitzzer_has_minimum_range](#)

omdurman-rules::src::range_effects::dervish_range_effects_monotone_non_increasing
omdurman-rules::src::range_effects::range_effects_first_range_max_effect_last_range_oor

§6.23 – Terrain defensive modifier

implemented

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Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

File	Symbol	Code Snippet
omdurman-rules/src/terrain_chart.rs:59 GH:omdurman-rules/src/terrain_chart.rs:59	defense_modifier	<pre>57 58 /// Convenience: get the defense modifier for a terrain type (rulebook §6.23, Terrain Effects Chart). 59 pub fn defense_modifier(terrain: Terrain) -> i16 { 60 terrain_effects_chart(terrain).defense_modifier 61 }</pre>
omdurman-rules/src/lib.rs:873 GH:omdurman-rules/src/lib.rs:873	FireModifier::Terrain	<pre>871 /// Negative modifier from the Terrain Effects Chart applied to the 872 /// defender's hex (§6.23). 873 Terrain(i16), 874 /// -2 thorn-hedge defensive modifier (§9.231). 875 ZaribaThornHedge,</pre>

Proven by: [omdurman-rules::src::lib::fire_modifier_keeps_roll_legal](#)

Covered

by tests: [omdurman-rules::src::terrain_chart::clear_terrain_no_bonus](#) [omdurman-rules::src::terrain_chart::building_gives_minus_3](#)
[omdurman-rules::src::terrain_chart::palm_grove_gives_minus_2](#) [omdurman-rules::src::terrain_chart::rough_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::swamp_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::hilltop_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::huts_movement_and_defense](#)
[omdurman-rules::src::terrain_chart::defense_modifier_convenience_matches_chart](#)
[omdurman-rules::src::terrain_chart::terrain_movement_costs_in_bounds](#)
[omdurman-rules::src::terrain_chart::terrain_defense_modifier_non_positive](#)

§6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

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All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:867 GH:omdurman-rules/src/lib.rs:867	AngloEgyptianDirectFire	<pre>865 pub enum FireModifier { 866 /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 867 AngloEgyptianDirectFire, 868 /// +1 brigade integrity, applied only if all four battalions fire at 869 /// the same enemy-occupied hex (§5.54, §6.24).</pre>
omdurman-rules/src/lib.rs:852	BrigadeIntegrity	<pre>850 /// stack contains all four battalions of a single Anglo-Egyptian brigade. 851 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]</pre>

GH:omdurman-rules/src/lib.rs:852		<pre> 852 pub enum BrigadeIntegrity { 853 None, 854 Integrated(BrigadeId), </pre>
omdurman-rules/src/lib.rs:967 GH:omdurman-rules/src/lib.rs:967	FireModifier::die_modifier	<pre> 965 impl MeleeModifier { 966 /// Return the numeric die-roll modifier for this 967 pub fn die_modifier(self) -> i16 { 968 match self { 969 MeleeModifier::DervishStandard => 2, </pre>

Proven by: [omdurman-rules::src::lib::die_roll_apply_modifier_is_total](#)

Covered by tests: [omdurman-rules::src::effects::tests::brigade_integrity_modifier_is_engine_derived](#)
[omdurman-rules::src::effects::tests::fire_modifiers_are_engine_derived_and_mismatches_rejected](#)

§6.41 – Direct Fire Subphase

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:304 GH:omdurman-rules/src/lib.rs:304	DirectFire	<pre> 302 pub enum FireSubPhase { 303 /// Direct fire (§6.41). Both sides participate in 304 DirectFire, 305 /// Anglo-Egyptian only: Maxim second fire + named- 306 MaximSecondAndHowitzer, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::fire_combat_elimimates_target](#)

§6.42 – Maxim Second Fire and Howitzer Fire Subphase

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:306 GH:omdurman-rules/src/lib.rs:306	MaximSecondAndHowitzer	<pre> 304 DirectFire, 305 /// Anglo-Egyptian only: Maxim second fire + named- 306 MaximSecondAndHowitzer, 307 } 308 </pre>
omdurman-types/src/lib.rs:967 GH:omdurman-types/src/lib.rs:967	fires_twice	<pre> 965 /// Maxim guns fire twice per turn -- once in the 966 Direct Fire Subphase and 967 /// again in the Maxim Second Fire Subphase 968 (rulebook §6.42). 969 pub fn fires_twice(self) -> bool { 970 matches!(self, UnitKind::Maxim { .. }) 971 } </pre>

Covered by tests: [omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10](#)
[omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7](#)
[omdurman-rules::src::effects::tests::advance_window_bridges_fire_subphase_and_closes_at_melee](#)
[omdurman-rules::src::effects::tests::fired_at_tracker_resets_at_maxim_subphase](#)

§6.51 – Leader Units

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:417 GH:omdurman-rules/src/lib.rs:417	BritishLeader	<pre> 415 /// to claim the Mahdi's Tomb (§9.14). 416 #[derive(Serialize, Deserialize, Clone, Copy, 417 pub enum BritishLeader { 418 Kitchener, 419 Gatacre, </pre>
omdurman-types/src/lib.rs:914 GH:omdurman-types/src/lib.rs:914	BritishLeader	<pre> 912 }, 913 /// Anglo-Egyptian leader (§6.51): movement only. 914 BritishLeader { movement: i32 }, 915 /// Wall-breach marker placed by artillery fire 916 Breech, </pre>
omdurman-types/src/lib.rs:955 GH:omdurman-types/src/lib.rs:955	has_combat_factors	<pre> 953 /// British leaders print a movement factor only 954 (\$6.51); other kinds carry 955 /// fire and/or melee factors. Markers carry no 956 stats. 957 pub fn has_combat_factors(self) -> bool { 958 !matches!(959 self, </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::zero_factor_is_none_not_zero](#)
[omdurman-rules::src::unit_profiles::kitchener_block_resolves_leaders_friendlys_camel_and_sudanese](#)
[omdurman-rules::src::unit_profiles::dervish_leader_sections_resolve_leader_and_retinue_per_cell](#)

§6.52 – Anglo-Egyptian Friendlys Brigade

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:592 GH:omdurman-rules/src/lib.rs:592	is_friendlys	<pre> 590 /// "Friendlys" units obey several special rules 591 (§5.21, §5.23, §6.52, 592 /// §9.14 victory conditions). 593 pub fn is_friendlys(&self) -> bool { 594 matches!(595 self, </pre>
omdurman-types/src/lib.rs:986 GH:omdurman-types/src/lib.rs:986	Friendlys	<pre> 984 /// Native volunteer brigade -- the Shaggyeh 985 (§6.52). Do not receive 986 /// brigade integrity (§5.54 enumerates only 987 British/Egyptian/Sudanese). 988 Friendlys, </pre>

Covered by tests: [omdurman-rules::src::unit_profiles::friendlys_counters_score_by_bank_not_as_leaders](#)
[omdurman-rules::src::effects::tests::friendlys_validate_and_resolve_on_dervish_table](#)

§6.53 – Royal Engineers demolition

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:563 GH:omdurman-rules/src/lib.rs:563	RoyalEngineers	<pre> 561 /// The Royal Engineers (§6.53) -- a *specific* 562 unit, not a class, so we 563 /// model it explicitly. 564 RoyalEngineers, 565 } </pre>

omdurman-rules/src/lib.rs:781 GH:omdurman-rules/src/lib.rs:781	demolishing	<pre> 779 /// Set when the Royal Engineers are committed to a 780 /// (\$6.53) -- neither offensive fire nor melee 781 /// allowed that turn. 782 pub demolishing: bool, 783 /// Set when a gunboat has lost its engines to a 784 /// river mine (\$10.12, roll 785 /// 5-7): it may no longer move under power and 786 /// instead drifts two hexes per </pre>
omdurman-rules/src/effects/effect.rs:175 GH:omdurman-rules/src/effects/effect.rs:175	Demolition	<pre> 173 174 /// Royal Engineers demolition (rulebook \$6.53). 175 Demolition { 176 unit_id: UnitId, 177 target: DemolitionTarget, </pre>
omdurman-rules/src/effects/setup.rs:35 GH:omdurman-rules/src/effects/setup.rs:35	apply_demolition	<pre> 33 /// resolution happens at end of turn via 34 /// ['apply_resolve_demolition'], which 35 /// checks the engineer is still adjacent and 36 /// undisrupted. 37 pub fn apply_demolition(38 state: &mut GameState, 39 unit_id: UnitId, </pre>
omdurman-rules/src/lib.rs:997 GH:omdurman-rules/src/lib.rs:997	DemolitionTarget	<pre> 995 /// disrupted or driven off. 996 #[derive(Serialize, Deserialize, Clone, Copy, 997 PartialEq, Eq, Debug)] 998 pub enum DemolitionTarget { 999 Fort(UnitId), 1000 WallHexside(HexsideRef), </pre>
omdurman-rules/src/effects/state.rs:2018 GH:omdurman-rules/src/effects/state.rs:2018	demolition_targets	<pre> 2016 /// the rules would accept. Empty when the unit 2017 /// doesn't exist or has no 2018 /// adjacent target. 2019 pub fn demolition_targets(&self, unit_id: UnitId) 2020 -> Vec<DemolitionTarget> { 2021 let Ok(unit) = self.unit_or_err(unit_id) else 2022 { 2023 return Vec::new(); </pre>

Covered by tests: [omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_disrupted](#)
[omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_moved_away](#)
[omdurman-rules::src::effects::tests::demolition_targets_finds_adjacent_fort_and_wall](#)

§6.54 – Forts

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:814 GH:omdurman-rules/src/lib.rs:814	ZocReason	<pre> 812 /// Used by the engine when answering "is this hex in 813 /// an enemy ZOC?". 814 #[derive(Serialize, Deserialize, Clone, Copy, 815 PartialEq, Eq, Debug)] 816 pub enum ZocReason { 817 /// Normal ZOC: any non-disrupted unit other than 818 /// an Anglo-Egyptian 819 /// leader (\$5.41) projects ZOC into each of its 820 /// six adjacent hexes. </pre>
omdurman-rules/src/lib.rs:821 GH:omdurman-rules/src/lib.rs:821	Fort	<pre> 819 GunboatVsGunboat, 820 /// Forts project ZOC out of, but not into, an 821 /// empty fort (\$5.44, §6.54). 822 Fort, 823 /// Walled-city ZOC: extends out through walls and 824 /// gates but not in, 825 /// across a breach in both directions (\$5.44). </pre>

omdurman-rules/src/lib.rs:769 GH:omdurman-rules/src/lib.rs:769	UnitState	<pre> 767 /// rather than one big enum. 768 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, Default)] 769 pub struct UnitState { 770 /// Reference table: "Disrupted units: no ZOC; may not move; may not fire 771 /// offensively or defensively; may not melee; are turned face up at the </pre>
omdurman-rules/src/lib.rs:912 GH:omdurman-rules/src/lib.rs:912	FireAttack	<pre> 910 /// modifiers (rulebook §6). 911 #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)] 912 pub struct FireAttack { 913 pub firing_player: Player, 914 pub phase: Phase, </pre>
omdurman-rules/src/lib.rs:927 GH:omdurman-rules/src/lib.rs:927	FireAttack::net_modifier	<pre> 925 impl FireAttack { 926 /// Sum of all fire modifiers applied to this attack (rulebook §6.24). 927 pub fn net_modifier(&self) -> i16 { 928 // Saturating fold rather than `sum()`: the modifier list is unbounded 929 // and arrives over the network, so a plain sum can overflow `i16`. </pre>

Covered by tests: [omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_enemy_fort](#)

§6.61 – Only artillery may fire at gunboats; 3+ to sink

implemented

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Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:476 GH:omdurman-rules/src/lib.rs:476	WeaponClass	<pre> 474 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 475)] 476 pub enum WeaponClass { 477 /// Dervish spears and swords -- no ranged fire at all. 478 Melee, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::rifles_may_not_sink_a_gunboat](#)

§6.62 – Only artillery may fire at forts; 2+ to destroy

implemented

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Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:476 GH:omdurman-rules/src/lib.rs:476	WeaponClass	<pre> 474 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug, strum::Display, 475)] 476 pub enum WeaponClass { 477 /// Dervish spears and swords -- no ranged fire at </pre>

		all. 478 Melee,
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Covered by tests: `omdurman-rules::src::effects::tests::artillery_destroys_fort_on_two_or_better_only`

§6.63 – Only artillery may breach wall hexsides; 2+ required

implemented

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Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

****6.64) Howitzer fire:**** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Subphase on the marchsheet for the impact hex. The designated target hex is hit on a roll of 7-10. (see manual for full text)

See also: §6.41, §6.42, §6.64

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:198 GH:omdurman-types/src/lib.rs:198	Breach	<pre>196 // Breach in a wall (artillery/§6.63 or Royal 197 // ways; LOS no longer blocked across the hexside. 198 Breach, 199 // Khor -- gully/wadi. ZOCs do not extend across 200 // combat may not cross (§6.82).</pre>
omdurman-rules/src/effects/effect.rs:278 GH:omdurman-rules/src/effects/effect.rs:278	ArtilleryBreachWall	<pre>276 // pre-rolled d10 used for the CRT lookup; range/ 277 // engine from the firers and `target`. 278 ArtilleryBreachWall { 279 firers: Vec<UnitId>, 280 target: HexsideRef,</pre>
omdurman-rules/src/effects/fire.rs:597 GH:omdurman-rules/src/effects/fire.rs:597	apply_artillery_breach	<pre>595 // artillery's CRT roll -- the rulebook specifies the 596 // threshold for both trigger styles. 597 pub fn apply_artillery_breach_wall(598 state: &mut GameState, 599 firers: &[UnitId],</pre>
omdurman-rules/src/effects/state.rs:1215 GH:omdurman-rules/src/effects/state.rs:1215	can_fire_at_wall	<pre>1213 // range band and resolving the CRT -- this method 1214 // firer at a time. 1215 pub fn can_fire_at_wall(1216 &self, 1217 firer: UnitId,</pre>

Covered by tests: `omdurman-rules::src::unit_profiles::breach_marker_cell_returns_none`

§6.64 – Howitzer fire

implemented

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File	Symbol	Code Snippet
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omdurman-rules/src/howitzer_scatter.rs:8 GH:omdurman-rules/src/howitzer_scatter.rs:8	ScatterHexDirection	<pre> 6 /// the diagram away from the firing player. 7 #[derive(serde::Serialize, serde::Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug)] 8 pub enum ScatterHexDirection { 9 UpperLeft, 10 UpperRight,</pre>
omdurman-rules/src/howitzer_scatter.rs:36 GH:omdurman-rules/src/howitzer_scatter.rs:36	howitzer_scatter	<pre> 34 /// onto a hex-grid offset oriented away from the firer (see 35 /// `GameState::howitzer_impact_hex`). 36 pub fn howitzer_scatter(impact_roll: DieRoll) -> ScatterHexDirection { 37 let table = crate::tables_data::scattergram_table(); 38 table</pre>
omdurman-rules/src/lib.rs:428 GH:omdurman-rules/src/lib.rs:428	GunboatId	<pre> 426 /// fire; "old" gunboats do not (rulebook §2.32). 427 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 428 pub enum GunboatId { 429 /// One of the five new-type named gunboats with howitzer capability. 430 Named(NamedGunboat),</pre>
omdurman-rules/src/effects/effect.rs:88 GH:omdurman-rules/src/effects/effect.rs:88	HowitzerFire	<pre> 86 /// - CRT result applied to units at impact hex (not the original target). 87 /// - Firers marked as fired. 88 HowitzerFire { 89 attack: FireAttack, 90 combat_results_table_roll: DieRoll,</pre>
omdurman-rules/src/effects/fire.rs:13 GH:omdurman-rules/src/effects/fire.rs:13	apply_howitzer_fire	<pre> 11 12 /// Validate and apply a howitzer fire attack (scatter path) (rulebook §6.64). 13 pub fn apply_howitzer_fire(14 state: &mut GameState, 15 attack: &FireAttack,</pre>
omdurman-rules/src/effects/state.rs:1030 GH:omdurman-rules/src/effects/state.rs:1030	can_fire_at	<pre> 1028 /// modifier in the [`FireAttack`] and is responsible for the LOS gate. 1029 /// (Howitzer fire ignores LOS entirely -- §6.64.) 1030 pub fn can_fire_at(1031 &self, 1032 firer: UnitId,</pre>

Covered by tests:

- [omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10](#)
- [omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7](#)
- [omdurman-rules::src::howitzer_scatter::howitzer_each_miss_gets_its_printed_hex](#)
- [omdurman-rules::src::unit_profiles::named_and_old_gunboats_resolve](#)
- [omdurman-rules::src::effects::tests::named_gunboat_has_howitzer](#)
- [omdurman-rules::src::effects::tests::named_gunboat_may_fire_howitzer_in_second_subphase](#)
- [omdurman-rules::src::effects::tests::named_gunboat_direct_fire_uses_artillery_weapon](#)
- [omdurman-rules::src::effects::tests::named_gunboat_no_howitzer_at_night](#)
- [omdurman-rules::src::effects::tests::dervish_gunboat_lacks_howitzer](#)

§6.81 – Moving player may fire with all capable units

implicit

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During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

§6.82 – Advance after combat (offensive fire)

implemented

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If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:162 GH:omdurman-rules/src/effects/effect.rs:162	AdvanceAfterCombat	<pre> 160 /// **Postconditions:** Unit position moved to `to`, `vacated_by_combat` 161 /// entry consumed. 162 AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 163 164 // -- Unit state changes ----- </pre>
omdurman-rules/src/effects/dispatch.rs:36 GH:omdurman-rules/src/effects/dispatch.rs:36	apply_advance_after_combat	<pre> 34 } 35 GameEffect::AdvanceAfterCombat { unit_id, to } => { 36 apply_advance_after_combat(state, *unit_id, *to) 37 } 38 GameEffect::RecoverUnit { unit_id } => apply_recover_unit(state, *unit_id), </pre>
omdurman-types/src/lib.rs:241 GH:omdurman-types/src/lib.rs:241	blocks_advance_after_combat	<pre> 239 240 /// Whether advance-after-combat may *not* cross this side (§6.82, §7.6). 241 pub fn blocks_advance_after_combat(self) -> bool { 242 matches!(243 self, </pre>
omdurman-rules/src/effects/state.rs:1922 GH:omdurman-rules/src/effects/state.rs:1922	can_advance_after_combat	<pre> 1920 /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 1921 /// Wall/khor hexside restrictions are not enforced (no hexside map data). 1922 pub fn can_advance_after_combat(&self, unit_id: UnitId, to: HexCoord) -> Result<(), RuleError> { 1923 let unit = self.unit_or_err(unit_id)?; 1924 // §6.7: there is no advance after combat as a result of defensive fire. </pre>

Proven by: [omdurman-rules::src::effects::advance_phase_is_atomic](#)

Covered by tests:

- [omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_wall_hexside](#)
- [omdurman-rules::src::effects::tests::can_advance_after_combat_rejects_khor_hexside](#)
- [omdurman-rules::src::effects::tests::advance_requires_combat_vacated_hex](#)
- [omdurman-rules::src::effects::tests::advance_requires_participation](#)
- [omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows](#)

8/8 implemented

§7 – Melee Phase

§7 – Melee Phase (chapter)

implemented

manual page unknown

Melee Phase

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:133 GH:omdurman-rules/src/effects/state.rs:133	PendingMelee	<pre>131 /// resolution after the reaction window is 132 deterministic and host-ordered (rulebook §7.5). 133 #[derive(Serialize, Deserialize, Clone, Debug)] 134 pub struct PendingMelee { 135 pub attack: MeleeAttack, 136 pub attacker_roll: DieRoll,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::declared_melee_blocks_phase_advance](#)

§7.1 – Melee strength printed on counter

implemented

manual page unknown

The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:110 GH:omdurman-rules/src/lib.rs:110	MeleeFactor	<pre>108 /// Every possible value from the annotated counter 109 set is a named variant. 110 #[derive(Serialize, Deserialize, Clone, Copy, 111 PartialEq, Eq, Hash, Debug, strum::Display)] 112 pub enum MeleeFactor { 113 One = 1, 114 Three = 3,</pre>
omdurman-rules/src/lib.rs:121 GH:omdurman-rules/src/lib.rs:121	MeleeFactor::sum	<pre>119 impl MeleeFactor { 120 /// Sum multiple melee factors (rulebook §7.1). 121 pub fn sum<'a>(factors: impl IntoIterator<Item = 122 &'a MeleeFactor>) -> u16 { 123 factors.into_iter().map(f f.value()).sum() 124 } 125 }</pre>
omdurman-types/src/lib.rs:938 GH:omdurman-types/src/lib.rs:938	may_be_melee_attacked	<pre>936 937 /// Gunboats neither attack nor are attacked in 938 melee (§7.1). 939 pub fn may_be_melee_attacked(self) -> bool { 940 !matches!(self, UnitKind::Gunboat { .. }) 941 }</pre>

Covered by tests: [omdurman-rules::src::effects::tests::reinforcement_rejected_onto_enemy_occupied_hex](#)

§7.2 – Melee adjacent only, not across wall hexsides

implemented

manual page unknown

Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:236 GH:omdurman-types/src/lib.rs:236	blocks_melee	<pre> 234 /// Whether melee may *not* be made across this side (§7.2). Gates and 235 /// breaches are passable to melee. 236 pub fn blocks_melee(self) -> bool { 237 matches!(self, HexsideKind::Wall HexsideKind::ZaribaThornHedge) 238 } </pre>
omdurman-rules/src/effects/state.rs:1305 GH:omdurman-rules/src/effects/state.rs:1305	can_melee	<pre> 1303 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 1304 /// thorn-hedge hexside blocks the attack (§7.2). 1305 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 1306 let unit = self.unit_or_err(attacker)?; 1307 </pre>

Covered by tests: [omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind](#)
[omdurman-rules::src::effects::tests::can_melee_rejects_wall_hexside](#)
[omdurman-rules::src::effects::tests::can_melee_rejects_thorn_hedge_hexside](#)
[omdurman-rules::src::effects::tests::can_melee_allows_gate_hexside](#)

§7.3 – Simultaneous melee combat

implemented

manual page unknown

Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:111 GH:omdurman-rules/src/effects/effect.rs:111	MeleeCombat	<pre> 109 /// - Winner may advance into vacated hex (§7.6). 110 /// - Victory points awarded for eliminations. 111 MeleeCombat { 112 attack: MeleeAttack, 113 attacker_roll: DieRoll, </pre>
omdurman-rules/src/effects/melee.rs:4 GH:omdurman-rules/src/effects/melee.rs:4	apply_melee_combat	<pre> 2 3 /// Apply a simultaneous melee combat between two adjacent hexes (rulebook §7). 4 pub fn apply_melee_combat(5 state: &mut GameState, 6 attack: &MeleeAttack, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::melee_resolves_simultaneously](#)

§7.4 – Who may melee attack / defend

implemented

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Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:927	may_melee_attack	<pre> 925 /// Rulebook §7.4 -- only infantry, cavalry, camel and Dervish leaders may 926 /// melee *attack*. All others (except gunboats) may melee *defend* (§7.1). </pre>

GH:omdurman-types/src/lib.rs:927		<pre> 927 pub fn may_melee_attack(self) -> bool { 928 matches!(929 self, </pre>
omdurman-types/src/lib.rs:864 GH:omdurman-types/src/lib.rs:864	UnitKind	<pre> 862 /// `Some(UnitKind::Marker)` or `None`. 863 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 864 pub enum UnitKind { 865 /// Foot infantry (§2.3): fire / melee / movement. 866 Infantry { </pre>
omdurman-types/src/lib.rs:713 GH:omdurman-types/src/lib.rs:713	DervishTribe	<pre> 711 Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display, strum::EnumIter, 712)] 713 pub enum DervishTribe { 714 Baggara, 715 Jaalin, </pre>
omdurman-rules/src/effects/state.rs:1305 GH:omdurman-rules/src/effects/state.rs:1305	can_melee	<pre> 1303 /// that may be melee-attacked (gunboats may not -- §7.1), and no wall or 1304 /// thorn-hedge hexside blocks the attack (§7.2). 1305 pub fn can_melee(&self, attacker: UnitId, defender_hex: HexCoord) -> Result<(), RuleError> { 1306 let unit = self.unit_or_err(attacker)?; 1307 </pre>

Covered by tests: `omdurman-rules::src::effects::tests::can_melee_gates_phase_adjacency_and_kind`

§7.5 – Cavalry/camel retreat before melee

implemented

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Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:148 GH:omdurman-rules/src/effects/effect.rs:148	RetreatBeforeMelee	<pre> 146 /// 147 /// **Postconditions:** Unit position moved to `to`. 148 RetreatBeforeMelee { unit_id: UnitId, to: HexCoord }, 149 150 /// An attacking unit advances into a hex vacated by combat (rulebook §6.82 </pre>
omdurman-rules/src/effects/dispatch.rs:33 GH:omdurman-rules/src/effects/dispatch.rs:33	apply_retreat_before_melee	<pre> 31 GameEffect::ResolveMelee => apply_resolve_melee(state), 32 GameEffect::RetreatBeforeMelee { unit_id, to } => { 33 apply_retreat_before_melee(state, *unit_id, *to) 34 } 35 GameEffect::AdvanceAfterCombat { unit_id, to } => { </pre>
omdurman-types/src/lib.rs:944 GH:omdurman-types/src/lib.rs:944	may_retreat_before_melee	<pre> 942 /// Cavalry and camel units may retreat two hexes from an infantry melee 943 /// attack (§7.5). 944 pub fn may_retreat_before_melee(self) -> bool { 945 matches!(self, UnitKind::Cavalry { .. } UnitKind::Camel { .. }) 946 } </pre>
omdurman-rules/src/lib.rs:187	HexDistance	<pre> 185 /// (rulebook §6.22, §7.5). 186 #[derive(Serialize, Deserialize, Clone, Copy, </pre>

GH:omdurman-rules/src/lib.rs:187		<pre>PartialEq, Eq, PartialOrd, Ord, Hash, Debug]] 187 pub struct HexDistance(u16); 188 189 impl HexDistance {</pre>
omdurman-rules/src/effects/state.rs:1853 GH:omdurman-rules/src/effects/state.rs:1853	can_retreat_before_melee	<pre>1851 /// two hexes away and empty. (Does not verify the 1852 /// the caller offers the retreat only in response 1853 /// to one.) 1854 pub fn can_retreat_before_melee(&self, unit_id: 1855 UnitId, to: HexCoord) -> Result<(), RuleError> { 1856 let unit = self.unit_or_err(unit_id)?; 1857 if !matches!(self.phase, Phase::Melee) {</pre>

Proven by: [omdurman-rules::src::effects::resolve_melee_is_atomic](#)

Covered by tests: [omdurman-rules::src::effects::tests::retreat_before_melee_only_cavalry_two_hexes](#)
[omdurman-rules::src::effects::tests::retreat_opens_window_only_when_hex_empties](#)
[omdurman-rules::src::effects::tests::rejected_resolve_melee_keeps_the_declaration](#)

§7.6 – Advance after melee

implemented

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player MUST advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units MUST advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:162 GH:omdurman-rules/src/effects/effect.rs:162	AdvanceAfterCombat	<pre>160 /// **Postconditions:** Unit position moved to 161 /// `to`, `vacated_by_combat` 162 /// entry consumed. 163 AdvanceAfterCombat { unit_id: UnitId, to: 164 HexCoord }, 165 166 // -- Unit state changes 167 -----</pre>
omdurman-rules/src/effects/dispatch.rs:36 GH:omdurman-rules/src/effects/dispatch.rs:36	apply_advance_after_combat	<pre>34 } 35 GameEffect::AdvanceAfterCombat { unit_id, to } => { 36 apply_advance_after_combat(state, *unit_id, 37 *to) 38 } 39 GameEffect::RecoverUnit { unit_id } => 40 apply_recover_unit(state, *unit_id),</pre>
omdurman-rules/src/effects/state.rs:1922 GH:omdurman-rules/src/effects/state.rs:1922	can_advance_after_combat	<pre>1920 /// player's unit, not artillery, adjacent to 1921 /// `to`, and `to` now empty. 1922 /// Wall/khor hexside restrictions are not 1923 /// enforced (no hexside map data). 1924 pub fn can_advance_after_combat(&self, unit_id: 1925 UnitId, to: HexCoord) -> Result<(), RuleError> { 1926 let unit = self.unit_or_err(unit_id)?; 1927 // §6.7: there is no advance after combat as a 1928 // result of defensive fire.</pre>

Proven by: [omdurman-rules::src::effects::advance_phase_is_atomic](#)

Covered by tests: [omdurman-rules::src::effects::tests::dervish_advance_after_melee_is_mandatory](#)
[omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows](#)

§7.7 – Melee modifiers

implemented

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To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:955 GH:omdurman-rules/src/lib.rs:955	MeleeModifier	<pre>953 954 #[derive(Serialize, Deserialize, Clone, Copy, 955 pub enum MeleeModifier { 956 /// +2 to all Dervish melee rolls (§7.7). 957 DervishStandard,</pre>
omdurman-rules/src/lib.rs:978 GH:omdurman-rules/src/lib.rs:978	MeleeAttack	<pre>976 /// A melee attack: simultaneous, both sides roll on 977 the Combat Results Table (§7.3, §7.7). 978 #[derive(Serialize, Deserialize, Clone, Debug, 979 PartialEq, Eq)] 980 pub struct MeleeAttack { 981 pub attacker_player: Player, 982 pub attacker_hex: HexCoord,</pre>
omdurman-rules/src/lib.rs:959 GH:omdurman-rules/src/lib.rs:959	MeleeModifier::Anglo-EgyptianStandard	<pre>957 DervishStandard, 958 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 959 AngloEgyptianStandard, 960 /// Inverted to -2 when Dervish units melee-attack 961 across a trench into 962 /// an entrenched defender (§9.232).</pre>
omdurman-rules/src/lib.rs:962 GH:omdurman-rules/src/lib.rs:962	MeleeModifier::DervishVsDefender	<pre>960 /// Inverted to -2 when Dervish units melee-attack 961 across a trench into 962 /// an entrenched defender (§9.232). 963 DervishVsTrenchedDefender, 964 }</pre>
omdurman-rules/src/lib.rs:957 GH:omdurman-rules/src/lib.rs:957	MeleeModifier::DervishStandard	<pre>955 pub enum MeleeModifier { 956 /// +2 to all Dervish melee rolls (§7.7). 957 DervishStandard, 958 /// +1 to all Anglo-Egyptian melee rolls (§7.7). 959 AngloEgyptianStandard,</pre>

Proven by: [omdurman-rules::src::lib::melee_modifier_keeps_roll_legal](#)

Covered by tests: [omdurman-rules::src::effects::tests::melee_resolves_simultaneously](#)
[omdurman-rules::src::effects::tests::melee_modifiers_are_engine_derived_and_mismatches_rejected](#)

2/3 implemented

§8 – Night Game Turns

§8 – Night Game Turns

descriptive

manual page unknown

Night Game Turns

§8.1 – Night effects

implemented

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The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

****8.2) Dervish Desertion Roll:**** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:156 GH:omdurman-rules/src/lib.rs:156	MovementAllowance::halve	<pre>154 impl MovementAllowance { 155 /// Night movement allowance = halved (round down) (rulebook §8.1, §5.11). 156 pub fn halve(self) -> Self { 157 let v = self.value() / 2; 158 MovementAllowance::try_from(v).expect("halved value always a named variant")</pre>
omdurman-types/src/lib.rs:778 GH:omdurman-types/src/lib.rs:778	DayNight	<pre>776 /// (rulebook §8.1). 777 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 778 pub enum DayNight { 779 Day, 780 Night,</pre>
omdurman-rules/src/range_effects.rs:68 GH:omdurman-rules/src/range_effects.rs:68	night_max_range	<pre>66 67 /// The halved maximum range at night (§8.1): round down, minimum 1. 68 pub fn night_max_range(weapon: WeaponClass, ae: bool) - > u8 { 69 let day = max_day_range(weapon, ae); 70 if day <= 1 { 1 } else { day / 2 }</pre>
omdurman-rules/src/lib.rs:1450 GH:omdurman-rules/src/lib.rs:1450	effective_movement_at_night	<pre>1448 /// Apply night-turn movement halving for Anglo- Egyptian units (§8.1): all 1449 /// Anglo-Egyptian movement allowances are halved (round down). 1450 pub fn effective_movement_at_night(1451 allowance: MovementAllowance, 1452 player: Player,</pre>

Proven by: [omdurman-rules::src::lib::movement_allowance_halve_never_panics](#)

Covered

by tests: `omdurman-rules::src::range_effects::night_max_ranges` `omdurman-rules::src::range_effects::night_max_ranges_remaining`
`omdurman-rules::src::range_effects::ae_rifle_at_night_matches_rulebook_example`
`omdurman-rules::src::range_effects::max_day_range_all_combos`
`omdurman-rules::src::effects::tests::night_movement_overlay_allowance_halved`

§8.2 – Dervish Desertion Roll

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:190 GH:omdurman-rules/src/effects/effect.rs:190	DervishDesertion	<pre>188 /// the effect. The Khalifa, gunboats, artillery, 189 and forts may not be 190 /// chosen. 191 DervishDesertion { 192 roll: DieRoll, 193 deserters: Vec<UnitId>,</pre>
omdurman-rules/src/turn_track.rs:63 GH:omdurman-rules/src/turn_track.rs:63	DervishDesertion	<pre>61 None, 62 /// Dervish desertion roll (§8.2) -- occurs on the 63 first night turn. 64 DervishDesertion, 65 /// Dervish reinforcements are available. 66 DervishReinforcements,</pre>
omdurman-rules/src/turn_track.rs:60 GH:omdurman-rules/src/turn_track.rs:60	TurnEvent	<pre>58 /// Special events that occur on specific turns 59 (rulebook §8.2, §9.112, §9.113). 60 #[derive(Serialize, Deserialize, Clone, Copy, 61 PartialEq, Eq, Debug)] 62 pub enum TurnEvent { 63 None, 64 /// Dervish desertion roll (§8.2) -- occurs on the 65 first night turn.</pre>

Covered

by

tests:

`omdurman-rules::src::effects::tests::desertion_count_is_floor_one_and_a_half`
`omdurman-rules::src::turn_track::desertion_on_first_night`
`omdurman-rules::src::effects::tests::desertion_roll_required_before_first_night_movement_ends`

 22/33 implemented

§9 – The Scenarios

§9 – The Scenarios

descriptive

manual page unknown

The Scenarios

§9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

Covered

by

tests:

omdurman-app::src::scenario_setup::campaign_has_no_fixed_placements omdurman-app::src::tests::scenario_maps_to_board

§9.2 – The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

Covered

by

tests:

omdurman-app::src::tests::start_game_scenario_selects_board omdurman-rules::src::effects::tests::remove_deployed_unit_happy_path

§9.3 – Bonus Game: Fall of Khartoum

out-of-scope

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Bonus Game: FALL OF KHARTOUM

Covered by tests: omdurman-rules::src::effects::tests::remove_deployed_unit_happy_path

§9.11 – Set Up (Campaign)

out-of-scope

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Set Up

§9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:11 GH:omdurman-rules/src/turn_track.rs:11	GameTime	<pre> 9 /// starts at one of these twelve times. 10 #[derive(Serialize, Deserialize, Clone, Copy, 11 pub enum GameTime { 12 SixAM, 13 EightAM,</pre>
omdurman-rules/src/turn_track.rs:47 GH:omdurman-rules/src/turn_track.rs:47	TurnEntry	<pre> 45 /// A single entry on the Turn Record Track (rulebook 46 #[derive(Serialize, Deserialize, Clone, Copy, 47 pub struct TurnEntry { 48 /// 1-based turn number. 49 pub turn: u8,</pre>
omdurman-rules/src/turn_track.rs:95 GH:omdurman-rules/src/turn_track.rs:95	CAMPAIGN_TURN_TRACK	<pre> 93 /// is turn 9, which carries the once-per-game Dervish 94 // the printed track prints "Dervish Desertion Roll / 95 pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [96 // Row 1, left->right: Sept 1, 6 am -> 8 pm, then 97 entry(1, SixAM, Day, TurnEvent::None),</pre>
omdurman-rules/src/turn_track.rs:242 GH:omdurman-rules/src/turn_track.rs:242	TurnLabel	<pre> 240 /// printed cell; `Blank` is for unused positions in 241 #[derive(Serialize, Deserialize, Debug, Clone, Copy, 242 pub enum TurnLabel { 243 Blank, 244 /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).</pre>

Covered

by tests: `omdurman-rules::src::turn_track::campaign_track_22_turns` `omdurman-rules::src::turn_track::desertion_on_first_night`
`omdurman-rules::src::turn_track::campaign_track_label_and_day_night_agree`
`omdurman-rules::src::turn_track::game_time_display_all_variants` `omdurman-rules::src::turn_track::turn_label_display`
`omdurman-rules::src::turn_track::turn_label_out_of_range_is_none`
`omdurman-rules::src::effects::tests::game_over_after_campaign_turns`

§9.13 – Special Rules (Campaign)

descriptive

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Special Rules

None.

§9.14 – Victory Conditions (Campaign)

implemented

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Victory Conditions

The Mahdi's Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi's Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

****Dervish Player receives:****

(see manual for full text)

See also: §5.21

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1104 GH:omdurman-rules/src/lib.rs:1104	VpSource	<pre>1102 /// the manual and the engine. 1103 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1104 pub enum VpSource { 1105 // ----- Anglo-Egyptian player receives: 1106 /// Mahdi's Tomb control at conclusion of play (§9.14).</pre>
omdurman-rules/src/lib.rs:1130 GH:omdurman-rules/src/lib.rs:1130	VpSource::points	<pre>1128 impl VpSource { 1129 /// VP awarded to `who_scores()` (rulebook §9.14). 1130 pub fn points(self) -> VictoryPoints { 1131 match self { 1132 VpSource::MahdisTomb => VictoryPoints::new(25),</pre>
omdurman-rules/src/lib.rs:1145 GH:omdurman-rules/src/lib.rs:1145	VpSource::who_scores	<pre>1143 1144 /// Which player receives these victory points (rulebook §9.14). 1145 pub fn who_scores(self) -> Player { 1146 match self { 1147 VpSource::MahdisTomb</pre>
omdurman-rules/src/lib.rs:1180 GH:omdurman-rules/src/lib.rs:1180	VictoryLedger	<pre>1178 /// Cumulative victory ledger for one scenario (rulebook §9.14). 1179 #[derive(Serialize, Deserialize, Clone, Debug, Default)] 1180 pub struct VictoryLedger { 1181 pub events: Vec<VpEvent>, 1182 }</pre>
omdurman-rules/src/lib.rs:1186 GH:omdurman-rules/src/lib.rs:1186	VpEvent	<pre>1184 /// A single victory-point scoring event (rulebook §9.14). 1185 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1186 pub struct VpEvent { 1187 pub turn: GameTurnIndex, 1188 pub source: VpSource,</pre>
omdurman-rules/src/lib.rs:1193 GH:omdurman-rules/src/lib.rs:1193	VictoryLedger::total_for	<pre>1191 impl VictoryLedger { 1192 /// Total victory points earned by a given player (rulebook §9.14). 1193 pub fn total_for(&self, player: Player) -> VictoryPoints { 1194 VictoryPoints(1195 self.events</pre>
omdurman-rules/src/lib.rs:1205 GH:omdurman-rules/src/lib.rs:1205	VictoryLedger::superiority	<pre>1203 /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 1204 /// (rulebook §9.14). 1205 pub fn superiority(&self) -> VictoryPoints { 1206 VictoryPoints(1207 self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value(),</pre>
omdurman-rules/src/lib.rs:1225 GH:omdurman-rules/src/lib.rs:1225	CampaignVictoryLevel	<pre>1223 /// Campaign-game victory levels (§9.14). 1224 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1225 pub enum CampaignVictoryLevel { 1226 Draw, 1227 Marginal(Player),</pre>
omdurman-rules/src/lib.rs:1234	CampaignVictoryLevel::f	<pre>1232 impl CampaignVictoryLevel { 1233 /// Assign a level from the net superiority (§9.14).</pre>

GH:omdurman-rules/src/lib.rs:1234		<pre>1234 pub fn from_superiority(s: VictoryPoints) -> Self 1235 { 1236 let net = s.0; // Positive -> Anglo-Egyptian thresholds: 15/30/50</pre>
omdurman-rules/src/effects/movement.rs:194 GH:omdurman-rules/src/effects/movement.rs:194	score_elimination	<pre>192 .find_unit(leader) 193 .is_some_and(u u.profile.identity.is_gordon()); 194 score_elimination(state, leader, ElimCause::Combat); 195 state.units.retain(u u.id != leader); 196 if is_gordon {</pre>
omdurman-rules/src/lib.rs:239 GH:omdurman-rules/src/lib.rs:239	VictoryPoints	<pre>237 /// (rulebook §9.14). 238 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug, Default)] 239 pub struct VictoryPoints(i32); 240 241 impl VictoryPoints {</pre>

Covered by tests: omdurman-rules::src::effects::tests::friendly_bank_scores_by_side
omdurman-rules::src::effects::tests::mahdis_tomb_not_scored_without_a_leader
omdurman-rules::src::effects::tests::mahdis_tomb_scores_for_anglo_egyptian_when_held
omdurman-rules::src::lib::vp_source_attributes

§9.21 – Set Up (Historical)

out-of-scope
manual page unknown

Set Up

§9.22 – Scenario Length (Historical)

implemented
manual page unknown

Scenario Length

6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:132 GH:omdurman-rules/src/turn_track.rs:132	HISTORICAL_TURN_TRACK	<pre>130 131 /// Historical scenario track (§9.22 -- 4 turns, Sept 2 6:00 am -> 12:00 pm). 132 pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [133 TurnEntry { 134 turn: 1,</pre>

Covered by tests: omdurman-rules::src::turn_track::historical_turn_all_four_turns

§9.23 – Special Rule: The Zariba

descriptive
manual page unknown

Special Rule: “The Zariba”

9.231) Thorn hedge hexsides: –2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

****9.232) Trench hexsides:**** -4 to die roll on all Dervish fire attacks vs. entrenched units only; -2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

§9.24 – Victory Conditions (Historical)

implemented

manual page unknown

Victory Conditions

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated) |
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player’s net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1264 GH:omdurman-rules/src/lib.rs:1264	HistoricalVictoryLevel	<pre>1262 /// draw"). 1263 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 1264 pub enum HistoricalVictoryLevel { 1265 Draw = 1, 1266 Marginal = 2,</pre>

Covered by tests:
[omdurman-rules::src::lib::historical_victory_level_for_dervish](#) [omdurman-rules::src::lib::historical_victory_level_for_anglo_egyptian](#)

§9.31 – Bonus game map

out-of-scope

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Only the small FALL OF KHARTOUM scenario map is used for this game.

§9.32 – Set Up (Bonus)

out-of-scope

manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

implemented

manual page unknown

Scenario Length

Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

File	Symbol	Code Snippet
omdurman-rules/src/turn_track.rs:172 GH:omdurman-rules/src/turn_track.rs:172	FALL_OF_KHARTOUM	<pre>170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,</pre>

Covered by tests: [omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night](#)
[omdurman-rules::src::turn_track::fall_of_khartoum_turns_3_to_8_are_day](#)

§9.34 – Special Rules (Bonus)

out-of-scope
manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

implemented
manual page unknown

Victory Conditions

Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.
- Dervish marginal: eliminate GORDON turn six.
- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.
- British decisive: GORDON survives end of turn eight.

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:1382 GH:omdurman-rules/src/lib.rs:1382	FoKVictoryLevel::resolve	<pre>1380 /// rulebook: GORDON dies turn 5 (tactical) with 24 Dervish losses (-2 1381 /// levels) nets a British marginal. 1382 pub fn resolve(gordon_died_turn: Option<u8>, ->="" 1383="" 1384="" base="Self::base(gordon_died_turn," base_idx="Self::LADDER</pre" dervish_lost:="" i16)="" let="" scenario_end_turn);="" scenario_end_turn:="" self="" u8,="" {="" =""></u8>,></pre>

Covered by tests: [omdurman-rules::src::lib::fok_victory_level_worked_example](#) [omdurman-rules::src::lib::fok_victory_level_gordon_died_early](#)
[omdurman-rules::src::lib::fok_victory_level_late_gordon_death](#) [omdurman-rules::src::lib::fok_victory_level_gordon_survived](#)

§9.111 – Dervish set up (Campaign)

implemented

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Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

****9.112) Dervish reinforcements:**** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:243 GH:omdurman-rules/src/effects/state.rs:243	setup_complete	<pre>241 /// currently shares the same "both sides deployed" gate; when a scenario 242 /// needs a different minimum, branch on `self.scenario` here. 243 pub fn setup_complete(&self) -> Result<(), RuleError> { 244 let has = player { 245 self.units</pre>

Covered by tests: [omdurman-rules::src::unit_profiles::hadendowa_first_cell_is_isa_zachneih](#)
[omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive](#)
[omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

§9.112 – Dervish reinforcements (Campaign)

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:182 GH:omdurman-rules/src/effects/effect.rs:182	PlaceReinforcements	<pre>180 // -- Reinforcement / placement ----- 181 /// Place reinforcements onto the map (rulebook §9.112, §9.113). 182 PlaceReinforcements(Vec<UnitPlacement>), 183 184 // -- Scenario-specific -----</pre>
omdurman-rules/src/effects/dispatch.rs:44 GH:omdurman-rules/src/effects/dispatch.rs:44	apply_place_reinforceme	<pre>42 GameEffect::Demolition { unit_id, target } => apply_demolition(state, *unit_id, *target), 43 GameEffect::PlaceReinforcements(placements) => { 44 apply_place_reinforcements(state, placements) 45 } 46 GameEffect::DervishDesertion { roll, deserters } => {</pre>
omdurman-rules/src/reinforcements.rs:69 GH:omdurman-rules/src/reinforcements.rs:69	dervish_campaign_sched	<pre>67 /// All reinforcements enter on the west edge, south of the Khor Shambat. 68 /// Each unit pays terrain cost of the hex it enters through. 69 pub fn dervish_campaign_schedule() -> ReinforcementSchedule {</pre>

		70 ReinforcementSchedule { 71 player: Player::Dervish,
omdurman-types/src/lib.rs:524 GH:omdurman-types/src/lib.rs:524	Location	522 /// Named map landmarks (rulebook mapsheet, §9.111, §9.113, §9.212 scenarios). 523 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 524 pub enum Location { 525 FortMakran, 526 NorthFort,
omdurman-types/src/lib.rs:604 GH:omdurman-types/src/lib.rs:604	SetupLetter	602 /// Each letter marks a specific hex where a Dervish leader is placed. 603 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 604 pub enum SetupLetter { 605 Y, 606 K,
omdurman-types/src/lib.rs:736 GH:omdurman-types/src/lib.rs:736	Faction	734 /// `Some(BrigadeId::friendlies())`. 735 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 736 pub enum Faction { 737 Dervish { 738 tribe: DervishTribe,

Covered by tests:

- [omdurman-rules::src::reinforcements::dervish_schedule_has_three_waves](#)
- [omdurman-rules::src::reinforcements::dervish_wave_one_has_baggaara_and_three_leaders](#)
- [omdurman-rules::src::reinforcements::dervish_wave_two_has_hadendowa](#)
- [omdurman-rules::src::reinforcements::dervish_wave_three_is_all_remaining](#)
- [omdurman-rules::src::reinforcements::wave_for_turn_returns_correct_wave](#)
- [omdurman-rules::src::effects::tests::campaign_reinforcements_gate_by_wave](#)
- [omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force](#)

§9.113 – Anglo-Egyptian reinforcements (Campaign)

implemented

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The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!
- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/reinforcements.rs:126 GH:omdurman-rules/src/reinforcements.rs:126	anglo_egyptian_campaig	124 /// - "Friendlies" enter via Abu Alim hut on the east bank (8 MP per unit). 125 /// - All other AE units enter via the Anglo-Egyptian Entrance Area (1 MP). 126 pub fn anglo_egyptian_campaign_schedule() -> ReinforcementSchedule { 127 let free_leaders = vec![128 CampaignLeader::British(BritishLeader::Kitchener),

Covered by tests: `omdurman-rules::src::reinforcements::anglo_egyptian_schedule_has_four_waves`
`omdurman-rules::src::reinforcements::anglo_egyptian_leaders_available_each_wave`
`omdurman-rules::src::reinforcements::anglo_egyptian_turn_four_is_all_remaining`
`omdurman-rules::src::effects::tests::campaign_reinforcement_cap_and_double_entry`
`omdurman-rules::src::effects::tests::campaign_gunboats_quota_three_per_turn`
`omdurman-rules::src::effects::tests::reinforcement_rejected_onto_enemy_occupied_hex`
`omdurman-rules::src::effects::tests::campaign_setup_rejects_non_initial_force`

§9.211 – Anglo-Egyptian set up first, moves second (Historical)

implemented

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:342 GH:omdurman-rules/src/effects/dispatch.rs:342	first_player	<pre>340 341 /// The player who moves first in a scenario (\$4, \$9.113, \$9.212, \$9.322). 342 pub fn first_player(scenario: Scenario) -> Player { 343 match scenario { 344 Scenario::Campaign => Player::AngloEgyptian,</pre>

Covered by tests: `omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units`

§9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

implemented

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.
- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
 - A: Ali Wad Helu
 - D: Sheik El Din
 - Y: Yakub
 - K: Khalifa Abdullah
 - S: Sherif
 - O: Osman Digna

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:353 GH:omdurman-rules/src/effects/state.rs:353	in_deployment_zone	<pre>351 /// plan / UI rather than this hex predicate. Documented, not silently 352 /// dropped. 353 pub fn in_deployment_zone(&self, player: Player, hex: HexCoord, is_boat: bool) -> bool { 354 // No board attached -> permissive (unit tests, unbound session). 355 if self.board.terrain.is_empty() {</pre>

omdurman-rules/src/lib.rs:387 GH:omdurman-rules/src/lib.rs:387	DervishLeader::setup_letter	<pre> 385 /// (§9.212): A→Ali Wad Helu, D→Sheik El Din, 386 /// S→Sherif, O→Osman Digna. Inverse of 387 /// [`dervish_leader_for_setup_letter`]. 388 pub fn setup_letter(self) -> SetupLetter { 389 match self { 390 DervishLeader::AliWadHelu => 391 SetupLetter::A, </pre>
omdurman-rules/src/lib.rs:403 GH:omdurman-rules/src/lib.rs:403	dervish_leader_for_setup_letter	<pre> 401 /// inherent impl here, so the mapping is a free 402 /// function -- the bijective 403 /// inverse of [`DervishLeader::setup_letter`]. 404 pub fn dervish_leader_for_setup_letter(letter: 405 SetupLetter) -> DervishLeader { 406 match letter { 407 SetupLetter::A => DervishLeader::AliWadHelu, </pre>

Covered by tests: [omdurman-rules::src::effects::tests::deploy_rejected_outside_zone](#)
[omdurman-rules::src::unit_profiles::embedded_leaders_resolve_from_their_host_section](#)
[omdurman-app::src::scenario_setup::historical_places_all_six_leaders_when_anchors_present](#)
[omdurman-app::src::scenario_setup::missing_anchor_is_reported_not_dropped_silently](#)
[omdurman-rules::src::effects::tests::setup_letter_dervish_leader_roundtrip](#)
[omdurman-rules::src::effects::tests::setup_letter_to_dervish_leader_known_values](#)
[omdurman-rules::src::effects::tests::historical_setup_rejects_not_in_play_units](#)

§9.231 – Thorn hedge hexsides

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:875 GH:omdurman-rules/src/lib.rs:875	ZaribaThornHedge	<pre> 873 Terrain(i16), 874 /// -2 thorn-hedge defensive modifier (§9.231). 875 ZaribaThornHedge, 876 /// -4 trench defensive modifier (§9.232). Only 877 /// applies vs. "entrenched" 878 /// units (those Nile-side of the trench hexside). </pre>
omdurman-types/src/lib.rs:206 GH:omdurman-types/src/lib.rs:206	ZaribaThornHedge	<pre> 204 Crest, 205 /// Historical-scenario thorn-hedge segment of the 206 /// Zariba (§9.231). 207 ZaribaThornHedge, 208 /// Historical-scenario trench segment of the 209 /// Zariba (§9.232). 210 ZaribaTrench, </pre>
omdurman-rules/src/board.rs:267 GH:omdurman-rules/src/board.rs:267	has_zariba_thorn_hedge	<pre> 265 /// hexside on its perimeter – i.e. whether the 266 /// ZaribaThornHedge modifier 267 /// applies (§9.231). 268 pub fn has_zariba_thorn_hedge(&self, hex: HexCoord) 269 -> bool { 270 for n in hex.neighbors() { 271 if let Some(kind) = 272 self.hexside_between(hex, n) </pre>

Covered by tests: [omdurman-rules::src::effects::tests::zariba_fire_penalties_apply_to_dervish_fire_only](#)

§9.232 – Trench hexsides

implemented

manual page unknown

File	Symbol	Code Snippet
omdurman-rules/src/lib.rs:876	ZaribaTrenchEntrenched	<pre> 876 /// -4 trench defensive modifier (§9.232). Only 877 /// applies vs. "entrenched" </pre>

lib.rs:878 GH:omdurman-rules/src/lib.rs:878		877 878 879 880	/// units (those Nile-side of the trench hexside). ZaribaTrenchEntrenched, }
omdurman-types/src/ lib.rs:208 GH:omdurman-types/src/lib.rs:208	ZaribaTrench	206 207 208 209 210	ZaribaThornHedge, /// Historical-scenario trench segment of the Zariba (§9.232). ZaribaTrench, /// One of the two end hexsides of a Zariba trench segment that connect to /// the Nile River (§9.233). Units may only enter/ leave the Zariba via

Proven by: `omdurman-rules::src::lib::melee_modifier_keeps_roll_legal`

Covered by tests: `omdurman-rules::src::effects::tests::trench_entrenched_units_take_trench_modifiers`

§9.233 – Zariba entry/exit costs

implemented

manual page unknown

Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

File	Symbol	Code Snippet
omdurman-types/src/lib.rs:258 GH:omdurman-types/src/lib.rs:258	blocks_movement	<pre>256 /// `omdurman-rules`). The trench *end* variants are therefore intentionally 257 /// not blocking. 258 pub fn blocks_movement(self) -> bool { 259 matches!(260 self,</pre>
omdurman-rules/src/board.rs:282 GH:omdurman-rules/src/board.rs:282	zariba_entry_surcharge	<pre>280 /// movement points to cross"). Returns 2 when the edge between `from` and 281 /// `to` is one of the two trench ends, else 0. 282 pub fn zariba_entry_surcharge(&self, from: HexCoord, to: HexCoord) -> i16 { 283 match self.hexside_between(from, to) { 284 Some(k) if k.is_zariba_trench_end() => 2,</pre>
omdurman-rules/src/effects/state.rs:848 GH:omdurman-rules/src/effects/state.rs:848	movement_cost_for	<pre>846 /// 847 /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 848 pub fn movement_cost_for(849 &self, 850 unit: &UnitPlacement,</pre>
omdurman-rules/src/effects/movement.rs:31 GH:omdurman-rules/src/effects/movement.rs:31	apply_move_unit	<pre>29 /// the true terrain cost (§5.11) and enforces gunboat upstream/downstream 30 /// allowances (§5.24); otherwise it falls back to the caller-supplied `cost`. 31 pub fn apply_move_unit(32 state: &mut GameState, 33 unit_id: UnitId,</pre>

Covered by tests: `omdurman-rules::src::effects::tests::zariba_end_hexside_costs_extra_mp`
`omdurman-rules::src::effects::tests::zariba_thorn_hedge_blocks_movement`

§9.321 – British set up (Bonus)

implemented

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The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
 - one Egyptian Battalion artillery unit
 - two British infantry units (represents Caucasian troops)
 - three Egyptian infantry units (represents Cairo “Bazouks”)
 - four Sudan infantry units (represents Sudanese blacks)
 - four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	FALL_OF_KHARTOUM	<pre>40 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 41 /// sprites). 42 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[43 FixedPlacement { 44 section: SectionName::BritishBoats,</pre>

Covered by tests:

- [omdurman-rules::src::effects::tests::confirm_ready_rejected_below_scenario_target](#)
- [omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)
- [omdurman-app::src::scenario_setup::fall_of_khartoum_reports_missing_palace](#)
- [omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#)
- [omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile](#)
- [omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile](#)
- [omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob](#)
- [omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine](#)
- [omdurman-rules::src::effects::tests::british_boats_named_vs_old_gunboat_detection](#)
- [omdurman-rules::src::effects::tests::fok_order_of_battle_british](#)
- [omdurman-rules::src::effects::tests::fok_order_of_battle_dervish](#)
- [omdurman-rules::src::effects::tests::fok_caps_bind_across_counter_variants](#)

§9.322 – Dervish enters turn one (Bonus)

implemented

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Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi’s combined west bank forces)
- 3 Dervish artillery units.

File	Symbol	Code Snippet
omdurman-rules/src/unit_profiles.rs:332 GH:omdurman-rules/src/unit_profiles.rs:332	ali_wad_helu	<pre>330 /// (3-6-9) -- the Degheim force of §9.322, printed on Baggara-backed 331 /// sprites. 332 pub fn ali_wad_helu(col: u32, row: u32) -> Option<Classification> { 333 match (col, row) { 334 (0, 0) => dervish_leader(DervishLeader::AliWadHelu),</pre>
omdurman-types/src/lib.rs:828 GH:omdurman-types/src/lib.rs:828	sections_for_picker	<pre>826 /// provides the 3 artillery, and HadendowaForts supplies the 827 /// Dervish-controlled North Fort sprite (§9.344). 828 pub fn sections_for_picker(self) -> Option<&'static</pre>

		<pre>[SectionName]> { 829 match self { 830 Scenario::Campaign Scenario::Historical => None,</pre>
--	--	---

Covered by tests: `omdurman-rules::src::unit_profiles::ali_wad_helu_block_resolves_leader_and_degelim_tribes`
`omdurman-rules::src::effects::tests::fok_setup_complete_requires_full_oob`
`omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile`
`omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks`
`omdurman-rules::src::effects::tests::fok_order_of_battle_dervish`
`omdurman-rules::src::effects::tests::fok_dervish_east_edge_on_diamond_board`

§9.341 – Turn 1 is always a night turn (Bonus)

`implemented`

manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

File	Symbol	Code Snippet
<code>omdurman-rules/src/turn_track.rs:172</code> GH:omdurman-rules/src/turn_track.rs:172	<code>FALL_OF_KHARTOUM</code>	<pre>170 /// (the rulebook fixes none); only `day_night` is rule-bearing (night halves 171 /// Anglo-Egyptian movement and ranges and bars howitzer fire, §8.1). 172 pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] = [173 TurnEntry { 174 turn: 1,</pre>

Covered by tests: `omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night`

§9.342 – All hexes are playable, including half hexes (Bonus)

`implemented`

manual page unknown

All hexes are playable, including hexes showing up half or less.

File	Symbol	Code Snippet
<code>omdurman-rules/src/board_data.rs:37</code> GH:omdurman-rules/src/board_data.rs:37	<code>fall_of_khartoum_map</code>	<pre>35 36 /// The Fall-of-Khartoum board (§9.3). 37 pub fn fall_of_khartoum_map_data() -> MapData { 38 FOK.get_or_init(parse(FOK RON)).clone() 39 }</pre>

Covered by tests: `omdurman-rules::src::effects::tests::fall_of_khartoum_board_excludes_no_hexes`

§9.343 – Both players use the Dervish Range Effects Table (Bonus)

`implemented`

manual page unknown

Both players must use the Dervish Range Effects Table.

File	Symbol	Code Snippet
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omdurman-rules/src/effects/fire.rs:47 GH:omdurman-rules/src/effects/fire.rs:47	range_band_for	<pre> 45 /// in FALL OF KHARTOUM *both* players use the Dervish Range Effects Table 46 /// (§9.343). 47 pub fn range_band_for(48 scenario: Scenario, 49 player: Player,</pre>
---	-----------------------	--

Covered by tests: [omdurman-rules::src::effects::tests::fok_both_players_use_dervish_range_table](#)

§9.344 – Dervish controls the North Fort (Bonus)

implemented

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

File	Symbol	Code Snippet
omdurman-rules/src/scenario_setup.rs:42 GH:omdurman-rules/src/scenario_setup.rs:42	FALL_OF_KHARTOUM	<pre> 40 /// North Fort uses a campaign HadendowaForts counter (one of the spare fort 41 /// sprites). 42 pub const FALL_OF_KHARTOUM_SETUP: &[FixedPlacement] = &[43 FixedPlacement { 44 section: SectionName::BritishBoats,</pre>
omdurman-rules/src/effects/state.rs:1387 GH:omdurman-rules/src/effects/state.rs:1387	hex_has_enemy_fort	<pre> 1385 /// may neither occupy an enemy fort nor advance after combat into one 1386 /// (forts are never captured -- only destroyed, \$6.62/\$6.53/\$7.6). 1387 pub fn hex_has_enemy_fort(&self, hex: HexCoord, mover: Player) -> bool { 1388 self.units.iter().any(u { 1389 u.position == hex</pre>

Covered by tests: [omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace](#)
[omdurman-app::src::scenario_setup::fall_of_khartoum_fort_landmarks_sit_at_the_correct_hexes](#)
[omdurman-app::src::scenario_setup::placement_done_gate_matches_by_identity_not_allocated_id](#)
[omdurman-rules::src::effects::tests::fok_order_of_battle_dervish](#)
[omdurman-types::src::lib::fok_picker_allowlist_has_dervish_entry_force_blocks](#)

§9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:1371 GH:omdurman-rules/src/effects/state.rs:1371	is_nile_mouth_crossing	<pre> 1369 /// must be named on the board, else this is `false` and the move falls 1370 /// through to the ordinary contiguous-Nile rules. 1371 pub fn is_nile_mouth_crossing(&self, from: HexCoord, to: HexCoord) -> bool { 1372 let white = self 1373 .board</pre>
omdurman-types/src/lib.rs:540 GH:omdurman-types/src/lib.rs:540	Location::WhiteNileMouth	<pre> 538 /// The off-board mouth of the White Nile branch (FALL OF KHARTOUM §9.345) -- 539 /// a British gunboat may cross to the Blue Nile mouth for 6 upstream MP. 540 WhiteNileMouth,</pre>

		541 /// The off-board mouth of the Blue Nile branch (FALL OF KHARTOUM §9.345). 542 BlueNileMouth,
--	--	---

Covered by tests: `omdurman-rules::src::effects::tests::fok_gunboat_crosses_between_nile_mouths`

§9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

`implemented`

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

File	Symbol	Code Snippet
omdurman-rules/src/effects/dispatch.rs:477 GH:omdurman-rules/src/effects/dispatch.rs:477	<code>check_gordon_palace</code>	475 /// after combat). Records the turn (which fixes the §9.35 victory level) and 476 /// ends the game. A no-op outside FoK, or once GORDON is already gone. 477 pub fn check_gordon_palace(state: &mut GameState) { 478 if state.scenario != Scenario::FallOfKhartoum state.gordon_eliminated_turn.is_some() { 479 return;
omdurman-rules/src/lib.rs:607 GH:omdurman-rules/src/lib.rs:607	<code>UnitIdentity::is_gordon</code>	605 /// Whether this is the GORDON leader unit (§9.32, §9.346) -- the immobile 606 /// palace defender whose elimination ends FALL OF KHARTOUM (§9.35). 607 pub fn is_gordon(&self) -> bool { 608 matches!(609 self,

Covered by tests: `omdurman-rules::src::unit_profiles::gordon_is_an_immobile_british_leader`
`omdurman-rules::src::effects::tests::gordon_survives_means_no_elimination`
`omdurman-app::src::scenario_setup::fall_of_khartoum_places_gordon_in_the_palace`

7/10 implemented

§10 – Optional Rules

§10 – Optional Rules

descriptive

manual page unknown

Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

§10.1 – River Mines

descriptive

manual page unknown

River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

§10.2 – River Chain

descriptive

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River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

§10.11 – Secretly record mine hexes

implemented

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Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/setup.rs:330 GH:omdurman-rules/src/effects/setup.rs:330	apply_place_mine	<pre>328 /// Lay a river mine during setup (§10.11). Validated by 329 /// [<code>GameState::can_place_mine`</code>]. 330 pub fn apply_place_mine(state: &mut GameState, hex: HexCoord) -> Result<(), RuleError> { 331 state.can_place_mine(hex)?; 332 state.mines.push(MinePlacement {</pre>
omdurman-rules/src/effects/state.rs:174 GH:omdurman-rules/src/effects/state.rs:174	GameState::mines	<pre>172 friendlies_transport: None, 173 optional_rules: Vec::new(), 174 mines: Vec::new(), 175 chain: None, 176 board: BoardInfo::default(),</pre>
omdurman-rules/src/	OptionalRule	<pre>314 /// two should be in play (rulebook §10). 315 #[derive(Serialize, Deserialize, Clone, Copy,</pre>

lib.rs:316 GH:omdurman-rules/src/lib.rs:316		PartialEq, Eq, Debug)) 316 pub enum OptionalRule { 317 RiverMines, 318 RiverChain,
--	--	---

Covered by tests: `omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup`
`omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule`

§10.12 – Mine resolution

implemented

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When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine's blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

File	Symbol	Code Snippet
omdurman-rules/src/effects/effect.rs:200 GH:omdurman-rules/src/effects/effect.rs:200	RiverMine	198 // -- Optional rules ----- 199 /// River mine resolution (rulebook §10.12). 200 RiverMine { 201 gunboat_id: UnitId, 202 hex: HexCoord,
omdurman-rules/src/effects/river.rs:146 GH:omdurman-rules/src/effects/river.rs:146	apply_river_mine	144 145 /// Apply a river-mine resolution (rulebook §10.12). 146 pub fn apply_river_mine(147 state: &mut GameState, 148 gunboat_id: UnitId,
omdurman-rules/src/lib.rs:1056 GH:omdurman-rules/src/lib.rs:1056	MineResult	1054 /// British gunboat enters a mined hex. 1055 #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1056 pub enum MineResult { 1057 /// Roll 1-4: no effect. 1058 NoEffect,

Covered by tests: `omdurman-rules::src::effects::tests::mine_fires_once_and_spare_dervish`

§10.13 – Mines consumed after both rolled for

implemented

manual page unknown

After both mines have been rolled for, no more are available.

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:146 GH:omdurman-rules/src/effects/river.rs:146	apply_river_mine	144 145 /// Apply a river-mine resolution (rulebook §10.12). 146 pub fn apply_river_mine(147 state: &mut GameState, 148 gunboat_id: UnitId,

Covered by tests: `omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish`

§10.14 – Dervish gunboats pass safely

implemented

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The Dervish player's gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:146 GH:omdurman-rules/src/effects/river.rs:146	apply_river_mine	<pre>144 145 /// Apply a river-mine resolution (rulebook §10.12). 146 pub fn apply_river_mine(147 state: &mut GameState, 148 gunboat_id: UnitId,</pre>

Covered by tests: `omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish`

§10.21 – Secretly record chain hexes

implemented

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Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:618 GH:omdurman-rules/src/effects/state.rs:618	can_place_chain	<pre>616 /// Read-only check of a river-chain placement in 617 setup (§10.21): Setup phase 618 /// and at most [<code>MAX_CHAIN_HEXES</code>] hexes. 619 pub fn can_place_chain(&self, hexes: &[HexCoord]) - 620 > Result<(), RuleError> { 621 self.require_setup_phase()?; 622 // Optional-rule gate: the chain exists only 623 when the River Chain option</pre>
omdurman-rules/src/effects/setup.rs:341 GH:omdurman-rules/src/effects/setup.rs:341	apply_place_chain	<pre>339 /// Lay (or replace) the river chain during setup 340 (§10.21). Validated by 341 /// [<code>GameState::can_place_chain</code>]. 342 pub fn apply_place_chain(state: &mut GameState, hexes: 343 &[HexCoord]) -> Result<(), RuleError> { 344 state.can_place_chain(hexes)?; 345 state.chain = Some(ChainPlacement {</pre>
omdurman-rules/src/effects/state.rs:1841 GH:omdurman-rules/src/effects/state.rs:1841	MAX_CHAIN_HEXES	<pre>1839 1840 /// Maximum contiguous Nile hexes the river chain may 1841 span (§10.21). 1842 pub const MAX_CHAIN_HEXES: usize = 4; 1843 //</pre>

Covered by tests: `omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup`
`omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule`

§10.22 – Gunboat stops on chained hex

implemented

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When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

File	Symbol	Code Snippet
omdurman-rules/src/effects/state.rs:887 GH:omdurman-rules/src/effects/state.rs:887	can_move_gunboat	<pre>885 /// upstream movement allowance is their maximum for that turn." Chained Nile 886 /// hexes stop the gunboat (§10.22). 887 pub fn can_move_gunboat(&self, 888 unit_id: UnitId,</pre>

Covered by tests: [omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk](#)

§10.23 – Sinking the chain

implemented

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No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

File	Symbol	Code Snippet
omdurman-rules/src/effects/river.rs:190 GH:omdurman-rules/src/effects/river.rs:190	apply_sink_chain	<pre>188 /// Sink the river chain (rulebook §10.23). Marks the placed chain cleared so it 189 /// no longer stops gunboats (§10.22). 190 pub fn apply_sink_chain(state: &mut GameState) -> Result<(), RuleError> { 191 match state.chain.as_mut() { Some(chain) if !chain.sunk => {</pre>

Covered by tests: [omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk](#)

0/1 implemented

§11 – Historical Notes

§11 – Historical Notes

descriptive

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Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,
(see manual for full text)

See also: §9.1, §9.2, §9.3

0/1 implemented

Credits

§Credits – Credits

[descriptive](#)

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 1/1 implemented

Combat Results Table (shared reference)

§CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

implemented

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File	Symbol	Code Snippet
omdurman-rules/src/combat_results_table.rs:10 GH:omdurman-rules/src/combat_results_table.rs:10	FireFactorRow	<pre>8 serde::Serialize, serde::Deserialize, Clone, Copy, 9 PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 10 pub enum FireFactorRow { 11 /// 1-5 factors 12 Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:46 GH:omdurman-rules/src/combat_results_table.rs:46	from_total	<pre>44 45 /// Determine which row a given total fire factor 46 pub fn from_total(total: u16) -> Self { 47 match total { 48 0..=5 => FireFactorRow::Row01to05,</pre>
omdurman-rules/src/combat_results_table.rs:87 GH:omdurman-rules/src/combat_results_table.rs:87	combat_results_table	<pre>85 /// D = `Disrupt` (1/2 of target units, round up) 86 /// 1..5 = `Eliminate(n)` (that many units removed) 87 pub fn combat_results_table(row: FireFactorRow, roll: 88 DieRoll) -> CombatResult { 89 let cells = crate::tables_data::crt_table() 90 .get(&row)</pre>
omdurman-rules/src/lib.rs:944 GH:omdurman-rules/src/lib.rs:944	CombatResult	<pre>942 /// * `--` -- no effect 943 #[derive(Serialize, Deserialize, Clone, Copy, 944 PartialEq, Eq, Debug)] 945 pub enum CombatResult { 946 NoEffect,</pre>

Covered by tests: [omdurman-rules::src::combat_results_table::ae_combat_results_table_lowest_is_no_effect](#)

[omdurman-rules::src::combat_results_table::ae_combat_results_table_highest_is_eliminate_5](#)

[omdurman-rules::src::combat_results_table::ae_combat_results_table_progresses_with_roll](#)

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[omdurman-rules::src::combat_results_table::fire_factor_row_index_sequential](#)

[omdurman-rules::src::combat_results_table::crt_all_rows_monotone_non_decreasing](#)

[omdurman-rules::src::combat_results_table::crt_cross_row_monotone_for_each_roll](#)

[omdurman-rules::src::combat_results_table::crt_lowest_row_is_worst_highest_row_is_best](#)

[omdurman-rules::src::combat_results_table::crt_eliminate_never_exceeds_5](#)

[omdurman-rules::src::combat_results_table::crt_every_cell_matches_the_table](#)

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| Reference – Charts and Tables

§Reference – Charts and Tables

out-of-scope

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